

Investigation into how to Increase Deep Learning Utilisation in Regulated Industries

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MSc in Artificial Intelligence
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Submitted by: Branden Millward

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Abstract

The use of Artificial Intelligence (AI) has seen a dramatic increase in adoption, driven by the ease of use and accessibility of Generative AI tools that utilise Deep Learning (DL) techniques and algorithms, such as Large Language Models (LLMs). However, many regulated industries still don't utilise this technology and capitalise on their potential benefits. Currently, the most used algorithms in the Financial Services industry for making financial decisions are tree-based and regression models. Even with many Deep Learning algorithms performing better in these areas. A major contributing factor to the slow adoption of Deep Learning is the perceived lack of transparency within these technologies, often referred to as "Black Box" techniques. This is currently in direct contradiction to the numerous regulations that this industry must adhere to. That all pushes and promotes the importance of transparency, accountability and governability when making decisions. This is where the emerging research into Explainable AI (XAI) focuses on methods to understand how AI works and its decision-making process in a way that humans can interpret. This thesis examines the benefits of incorporating Deep Learning into the decision-making process, utilising Explainable AI techniques to comply with various regulations, with a particular focus on the FCA. The thesis demonstrates both the benefits of using Deep Learning in the decision-making process and how companies can comply with the FCA regulations. A list of recommendations and guidelines has been produced and submitted to the FCA to assist in future iterations of the regulations. Based on the author's industry experience working with financial services and research conducted as part of this investigation, a focus on demonstrating the benefits of Deep Learning and how it can comply with FCA regulations is novel work.

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I acknowledge that this work is my own, and I used Grammarly (Grammarly, 2025) to proofread my final draft only.

Chapter 1

Introduction

1.1 Problem Statement

This investigation aims to address the problem of the low adoption rate of Deep Learning techniques despite their proven superiority over traditional machine learning methods. Research has shown that Deep Learning Models performance “completely surpasses traditional methods” (Laqtib, Yassini, & Hasnaoui, 2019). However, the current adoption rate of Deep Learning techniques remains low, with Neural Networks (both Deep and Shallow) accounting for only ~12.8%. In comparison, Category Boosted methods made up 21% of the studies, and Traditional methods made up the majority, with a usage rate of 61.6% (Noriega, Rivera, & Herrera). This issue of low adoption rate is not confined to a single industry. However, regulated industries face particular scrutiny when adopting this technology due to specific requirements set out by their regulators. One of the main contributing factors to the lack of adoption in these industries is the vagueness of the regulations and a limited understanding of the options available in this emerging field that would enable businesses to remain compliant.

The goal of this investigation is to develop a list of recommendations, along with practical examples, that aim to address the vagueness of regulations and present options that can be adopted, enabling businesses to comply with their regulatory obligations. The desired impact of these outputs is threefold: firstly, to drive commercial benefits by encouraging companies to adopt techniques that enhance performance. Secondly, for consumers, the improved performance of the business should lead to a better customer experience, due to more accurate decision-making and analysis, which in turn produces improved outcomes. Lastly, these outputs and recommendations can enhance the clarity of regulations and serve as examples of “best practices” for regulators to share with the companies they oversee. With these impacts in mind, the output of this investigation should provide a significant net positive for all parties involved and impacted by these techniques. Based on the

problem statement identified, this investigation will next provide background and context on key areas impacting this problem, before examining potential solutions and producing the outputs.

1.2 Context and Background

1.2.1 What is Deep Learning (DL)

Deep Learning (DL) is a sub-category of Machine Learning (ML), which by extension is a subfield of Artificial Intelligence (AI). Deep Learning originated from Artificial Neural Networks (ANNs). These Artificial Neural Networks (ANNs) consist of a minimum of three layers: an input layer, a hidden layer, and an output layer. With each layer being made up of nodes (Artificial Neuron) “Artificial Neuron is basically an engineering approach of biological neuron” (Sharma, Rai, & Dev, 2012). These layers send values from each node to the nodes of the next layer by imitating the electrical signals sent through a synapse (see Figure 1), which forms the basic structure of a Multilayered Artificial Neural Network.

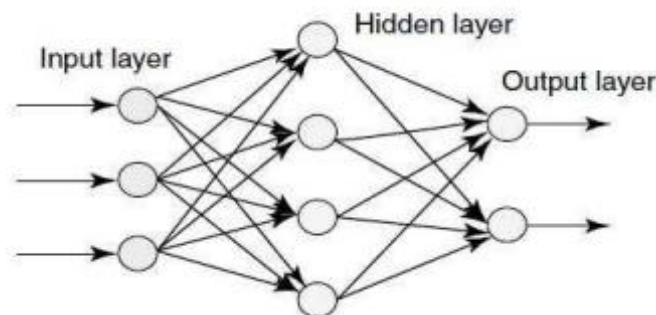


Figure 1 Multilayered Artificial Neural Network (Sharma, Rai, & Dev, 2012)

However, while Machine Learning (ML) and Deep Learning (DL) both focus on similar problems to solve and focus on the same three main learning types, these being Supervised Learning where “The operator provides the machine learning algorithm with a known dataset that includes desired inputs and outputs” (Wakefield, 2023) generally used for classification; Unsupervised Learning “There is no answer key or human operator to provide instruction. Instead, the machine determines the correlations and relationships by analysing available data” (Wakefield, 2023) these problems generally look at similarities between data to cluster/ group similar data points and lastly Semi-Supervised or Hybrid Learning where there is a mixture of both tagged and untagged where they can learn from the labelled data to identify and label the unlabelled data.

While Machine Learning (ML) has a variety of solutions that accomplish these tasks ranging from regression models to Shallow Neural Networks (SNNs), Deep learning focuses on the use of Neural Networks (NNs) with common approaches being, Multi-layer Perceptron (MLP), Convolutional Neural Network (CNN) and Recurrent Neural Network (RNN) for Supervised Learning, Generative Adversarial Network (GAN) for Unsupervised Learning and a combination of techniques for Semi-supervised/Hybrid Learning.

So what differentiates Deep Learning Neural Networks (NNs) is the number of hidden node layers, while there is no set rules it is commonly considered that Neural Networks up to 3 hidden layers are considered Shallow Neural Networks as shown in “In simple ANNs, there exists an input layer between zero and three hidden layers and an output layer” (Choi, Coyner, Kalpathy-Cramer, Chiang, & Campbell, 2020) and “very simple neural network architectures containing one or two layers” (Aggarwal, 2018). When a Neural Network has more than three hidden layers, it is generally considered Deep and is illustrated in Figure 2.

The hidden nature of these layers is what is commonly referred to as a “black box”, “the black box nature of the system means we can’t trace the system’s thought process and see why it made this decision”, (Rawashdeh, 2023), this is a key area that this dissertation is looking to resolve as it is seen as a major blocker to the adoption of Deep Learning and is seen as a disadvantage of Deep Learning.

1.2.2 The Benefits and Challenges of Deep Learning

There are various benefits and challenges associated with using Deep Learning that should be considered when adopting these techniques.

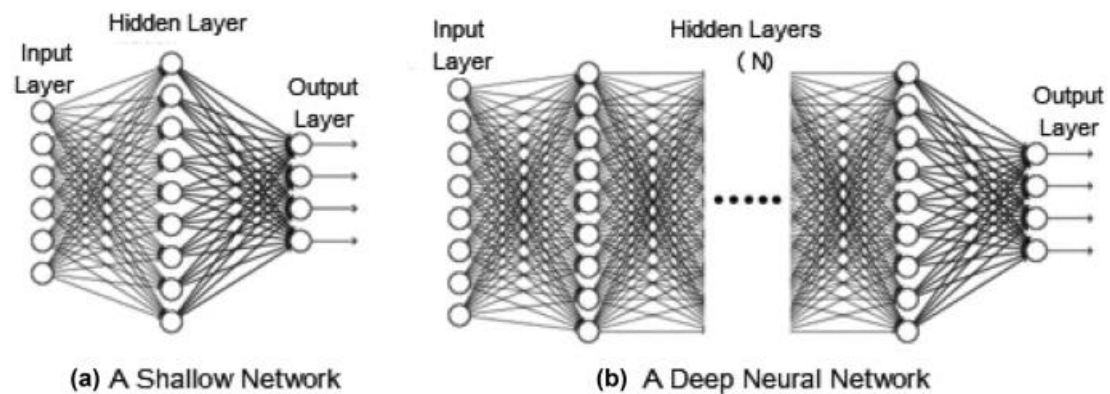


Figure 2 Representative examples of Shallow and Deep Neural Networks (Sarker, 2021)

1.2.2.1 Benefits

One of the primary driving factors for the adoption of Deep Learning is the increased

accuracy and performance it can achieve compared to other methods. An example of this was demonstrated by comparing a Support Vector Machine (SVM), a machine learning classification model, with a Convolutional Neural Network (CNN), a type of Deep Learning model. The 2 models were trained and tested on the same data to classify a variety of images and image sizes. The results are presented in Table 1 and Figure 3. Illustrate that as the picture size increases, the disparity between the 2 models increases, with the deep learning model dramatically increasing in accuracy from

Picture size	Number of categories	SVM classification Accuracy	CNN classification Accuracy
64x64	10	0.62	0.71
128x128	10	0.64	0.74
256x256	10	0.61	0.95

Table 1 Accuracy comparison of two algorithms under different picture sizes (Wang, Fan, & Wang, 2021)

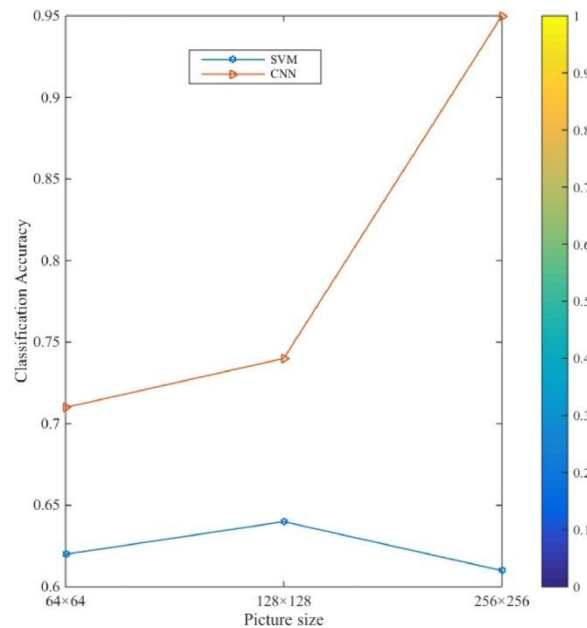


Figure 3 Accuracy comparison of two algorithms under different picture sizes (Wang, Fan, & Wang, 2021)

Another contributing factor to the increased use of Deep Learning can be linked to increased volumes of data and the complexity of that data that is everywhere in our current society, and is expected to continue to grow, as shown in Figure 4, this increase to the amount of data that companies are looking to handle and process has led to the pursuit of new technologies and has led to a rise in Deep Learning adoption, "The ability of deep learning to achieve high-level features from a massive amount of input data, referred to as feature engineering, distinguishes it from machine learning.

As a result, deep learning is gaining popularity with innovative applications” (Ahmed, et al., 2023).

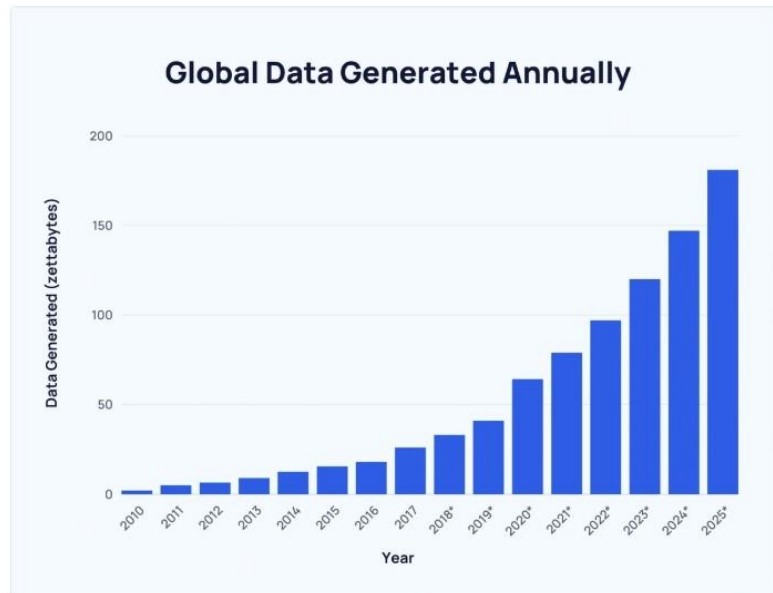


Figure 4 Global Data Generated Annually (Duarte, 2025)

1.2.2.2 Challenges

While there are benefits to using Deep Learning techniques, they come with their own sets of challenges that can act as an initial barrier to utilisation. As mentioned in the above benefits section, Deep Learning has an advantage over other techniques when handling large amounts of data and data that can be less structured than what is typically required of many Machine Learning models. This comes at a high cost, with examples of widely used Large Language Models (LLMs), which are an example of Deep Learning, costing in the tens to hundreds of millions of dollars, depending on the size and scale of the model. “According to Epoch AI, these can make up 29% to 49% of the final price. ChatGPT-4, the latest edition, had a technical creation cost of \$41 million to \$78 million, according to the source. Sam Altman, CEO of OpenAI” (Buchholz, 2024).

Additionally, while Deep Learning can suffer from similar challenges as other modelling techniques, such as overfitting the model to training datasets, concerns around what is input into the model, like using biased datasets to train the model or data security and privacy concerns, or in the case of regulated industries there may be protected characteristics that cannot be used to make decisions on or target based on. This challenge is exacerbated for Deep Learning models and projects due to the

nature of the hidden layers. The volume of hidden layers and the nature of these layers lead to a lack of interpretability and explainability in the models, as well as in the decision-making process or output generation. This has led to many Deep Learning techniques being labelled “Black-box Models”. This refers to “Deep Neural Networks with thousands or even millions of parameters (weights), are considered black boxes because the model’s behaviour cannot be comprehended, even when one is able to see its structure and weights.” (Perteneder, 2021).

This dissertation will address the second challenge by making Deep Learning a viable option for companies operating within regulated industries to utilise while complying with regulations. The objectives and scope sections of this dissertation will illustrate this focus.

1.2.3 What is a Regulated Industry

Regulations are laws or guidelines typically established by the government or its agencies. These regulators can be either the government, or a separate entity set up by and accountable to the government. In these cases, “regulation is the use of rules, incentives and penalties to change the behaviour of individuals or organisations. While rules may be set out in law, this is not always the case. Regulation involves not only setting standards, but monitoring performance against them and enforcing compliance.” (Bishop, 2024).

Another form of regulation is Self-Regulation, where companies within an industry come together and adopt their own set of rules and standards that are enforced similarly to those of laws set out by the government. One of the main reasons a company would adopt these guidelines and rules is the monetary impact on its business, as these regulations are often seen as a pillar of trust by consumers, offering a level of protection that indicates the company is reputable and trustworthy.

Regulators generally work with companies in 3 ways: a “Directive” approach that utilises its power over the industry to enforce its views; an “Engaged” approach that promotes companies to have an voice their views to the regulator but the decision would still sit with the regulator and finally a “Collaborative” approach which encourages companies to work with the regulator to form new regulations that are in the best interests of both the companies and consumers such as the pre-approved tag when looking at loans and credit cards, several companies have now pushed that to show a balance-transfer card as pre-approved you must guarantee the credit limit will be sufficient to allow for the balance transfer “We’ll let you know if your credit limit covers the amount you want to transfer, before you apply.”(Capital One, 2025).

The primary regulator that will be the focus of this investigation is the Financial Conduct Authority (FCA), which regulates Financial Services in the UK. However, there are many regulatory bodies across various industries, including Ofgem (Office of Gas and Electricity Markets) for energy and Ofwat (Water Services Regulatory Authority) for water. I have included a more comprehensive list of regulators in Appendix D.

1.2.4 What are the regulations and restrictions on using Deep Learning in a regulated Industry

Many of the regulators listed above do not give clear rules on the use of Deep Learning. However, they give some guidance on the use of Artificial Intelligence and Machine Learning more generally. With the UK government taking a “A pro-innovation approach to AI regulation” (The Secretary of State for Science, Innovation and Technology, 2023), this guided regulators to support the adoption of AI while voicing concerns “that certain unforeseen risks, gaps, and shortcomings might require further guidance or intervention in the future” (Cooley, 2024).

There has been a framework laid out for regulators with 5 core principles for using AI responsibly, these being: Safety, Security and Robustness; Appropriate Transparency and Explainability; Fairness; Accountability and Governance; Contestability and Redress (Gallo & Nair, 2024).

While these core principles are widely accepted and agreed by regulators across sectors, this poses a major problem to the adoption of Deep Learning techniques in their standard setup. Methods for accomplishing this are the core of this investigation, ranging from the preparation of the data used and accessed by the model to the explainability and ability to scrutinise the model and the decisions it outputs.

1.2.5 How can Deep Learning techniques be explained

Interpreting AI models and their outputs is a topic that dates back to the 1970s. However, the more commonly used Explainable AI (XAI) was defined in 2004 by (Lent , Fisher, & Mancuso, 2004). The purpose of Explainable AI (XAI) aligns closely with several core principles established by the government regarding the use of AI. This includes the ability to clearly explain to users the initial inputs to the AI algorithms, the interaction and reasoning processes, and the resulting output and actions, in a format that is easily understandable.

Explainable AI (XAI) has many subsections and methods that will be covered in more detail throughout this investigation. However, at a high level, Explainable AI (XAI) can be split into 3 categories: explaining the data, explaining the predictions and explaining the method or algorithm. This is generally complete through either utilising models to be interpretable and easy to explain, or using a secondary simple model to approximate and explain how a complex model made choices. This is the method commonly associated with Deep Learning and will be a big focus of this investigation.

By having these explanations, the hope is that models will adhere to many of the core frameworks, including demonstrating appropriate transparency and explainability, allowing users to illustrate how the model in use is fair and allowing regulators or other concerned parties to keep users accountable and be able to contest choices because

the data, methods and algorithm will be understood and clear.

1.3 Objectives

The overall goal of this investigation is to increase the adoption rate of Deep Learning techniques within regulated industries. While all data projects must broadly adhere to the framework laid out in (Gallo & Nair, 2024). The main barrier to the adoption of Deep Learning is believed to be a lack of understanding and trust in what data is being used and how it is being utilised. The focus of this investigation will be on the “Black Box” perception of Deep Learning. These perceptions are influenced by how Transparent and Explainable a model is, making it difficult for regulators to ensure Fairness, Accountability, Governance, Contestability, and allowing for Redress. This author believes that the following objectives can alleviate this blocker.

1.3.1 Objective 1

Demonstrate that Deep Learning techniques are worth utilising by showing increased performance over traditional models. All model comparisons will be conducted using the same datasets to ensure consistency in performance evaluations.

1.3.2 Objective 2

Produce examples of Deep Learning models that have been explained in a clear manner, which include the input choices, and utilise Explainable AI (XAI) techniques to ensure that the use of Deep Learning techniques adheres to the 5 key points laid out in the framework in (Gallo & Nair, 2024).

1.3.3 Objective 3

Produce a set of recommendations and guidance that can be used to both amend the regulations of the FCA or other regulators, and be utilised by businesses to comply with the regulations, while utilising deep learning techniques

The author's belief is that by completing these objectives, regulated industries will adopt and utilise Deep Learning methods more, improving business and customer outcomes, as the outputs of this investigation will:

- Illustrate the benefits of using Deep Learning, as outlined in Objective 1, and how this can impact a business,

- Providing practical examples of how businesses can unlock these benefits while adhering to regulations through Objective 2, with explanations of the models that have been used to attain the improved performance.
- Lastly, producing a list of recommendations as stated in Objective 3, for both regulators and businesses to utilise, ensuring a clear understanding of best practices and standards.

1.4 Scope

1.4.1 Industry

While data types and use cases vary across industries, many general concerns and solutions are transferable. An example of where industries may differ is in healthcare, where factors like gender and race affect health conditions but aren't relevant for credit risk assessments. This influences the input data for model building, yet common use cases like Image Classification, Structured Data Classification, Sentiment Analysis, or Time Series Forecasting span multiple industries.

This investigation focuses on the financial services industry as a representative of a regulated industry, using a credit risk dataset and adhering to regulations set by the Financial Conduct Authority (FCA) in the UK. However, references will be made to other industries in terms of their current usage and a comparison of regulations across industries. Many of the tools and recommendations will be transferable across industries.

1.4.2 Model target

The models to be built will aim to classify whether an account will “charge-off”, meaning the lender has written off the debt as a loss. The predictions of Charge-offs are crucial in financial services industry, providing insights into the risk levels of portfolios and can indicate wider economic events (Kim & Son, 2024). This significance is why this metric is the focus of the investigation.

1.4.3 Commercial potential of this investigation

This will have a significant commercial potential in regulated industries as it will open up the opportunity to utilise these techniques without regulatory ramifications. As such, they can benefit from the enhanced performance. Using the example of financial services and the model data type selected for this investigation, “banks with high loan defaults on their books can face insolvency if they are not able to recover their bad debts. Earnings of shareholders may suffer as a result of loan defaults.” (Asalanka, Niswah, & Kumara, 2023).

Chapter 2

Literature and Technology Survey

2.1 The Use of Deep Learning in Regulated Industries

2.1.1 Reasons why Explainable AI is important in regulated industries

Throughout the research conducted, there is an acknowledgement that the general technology encompassed by the umbrella term Artificial Intelligence is here to stay and is already being utilised in many aspects of life. In (Rashid & Kausik, 2024), it provides a good overview of how AI is being utilised across a variety of industries, additionally highlighting some of the potential future applications that these industries could benefit from, with applications varying from self-driving cars for the automotive industry, disease diagnosis in the healthcare industry and risk management in the financial industry, which was selected as the focus example for this investigation. It goes on to say that these advancements are contingent on the further advancement of “deep learning, neural networks, and natural language” (Rashid & Kausik, 2024). This sentiment is reiterated in both (The Secretary of State for Science, Innovation and Technology, 2023) and (Directorate-General for Communications Networks, Content and Technology, 2025), which highlight that the UK government and many other governments across the world have shown a clear stance that AI is a technology that will be pivotal to developed for countries, companies and peoples general lives going forward. While these governments have instructed regulators to take a pro-AI approach to regulation, to incentivise advancements, they have also made clear that when using this technology, it should not be at the detriment of transparency, safety, and security of the decisions, data, and uses for which the AI is utilised. In (Pantanowitz, et al., 2024), they speak about the importance of regulation being both flexible enough to not stifle growth but also ensure safe and effective adoption while promoting trust by the public.

In (Sarker, 2021) this advancement has been referred to as the "Fourth Industrial Revolution ... or Industry 4.0" (Sarker, 2021), it covers the importance of Deep Learning for future development and states how it is the main focus of this next revolution for industry, giving a clear overview of how Deep Learning is capable of benefiting a variety of industries however it stresses the challenges of Deep Learning and reiterates the opinion that Deep Learning is a “black-box solution” (Sarker, 2021), stating that the resolution of this challenge could lead to the next generation of Deep Learning advancements.

As part of (Mienye, Jere, Obaido, Mienye, & Aruleba, 2024), a list of potential challenges that could be faced when using Deep Learning within the Finance industry, highlighting again issues that face all modelling techniques, such as “Data Quality and Availability”, but similarly to (Sarker, 2021) it was raised that both “ethical and regulatory concerns” (Mienye, Jere, Obaido, Mienye, & Aruleba, 2024), along with the interpretability of Deep Learning Models. These challenges both highlighted the need for transparency of what and how decisions are made based on AI models, especially Deep Learning models, however, it was highlighted that XAI could be a solution to both interpretability and explainability issues of Deep Learning models and, by extension, go towards addressing ethical and regulatory concerns.

While the above illustrates the need and desire for explainable and interpretable models and decision making when using them within regulated industries, these terms are often used interchangeably in various literatures. This is usually due to the lack of a standardised measurement for these terms. Some papers have attempted to define interpretability as “the ability to explain or to present in understandable terms to a human” (Doshi-Velez & Kim, 2017). Whilst (Miller, 2019) defines interpretability as “the degree to which an observer can understand the cause of a decision” (Miller, 2019). Based on these definitions and similar, interpretable models can be explained to a human so they can understand how outputs are derived from inputs, but without giving specifics of each decision. Whilst Explainability is defined as “The Explainability property of a model includes the ability to provide detailed and understandable reasoning for specific decisions” (Frasca, Torre, Pravettoni, & Cutica, 2024). While a model can be interpreted and understood at a conceptual level, this doesn’t require a human to comprehend its inner workings. As such, a model can be interpretable without being explainable, and vice versa. However, this would not be adequate when working in regulated industries that rely on both the interpretability and explainability of a model. (Linardatos, Papastefanopoulos, & Kotsiantis, 2020), created a taxonomy of the various methods that can be used to interpret Machine Learning models, split into 4 categories: explaining complex black-box models; creating white-box models; promoting fairness and restricting the existence of discrimination; and analysing the sensitivity of model predictions.

With examples of “white-box models” being regression models and tree-based models, which are the models most commonly used within regulated industries. Deep Learning methods, as mentioned various times throughout this investigation and in other literature, would fall within the “black-box models” category.

An investigation was conducted to establish a framework for the application of AI in the financial services industry (Adams & Hagrais, 2020). This sets out explainability as the most important principle for using AI in financial services. Seen in Figure 5.

Investigation into how to Increase Deep Learning Utilisation in Regulated Industries

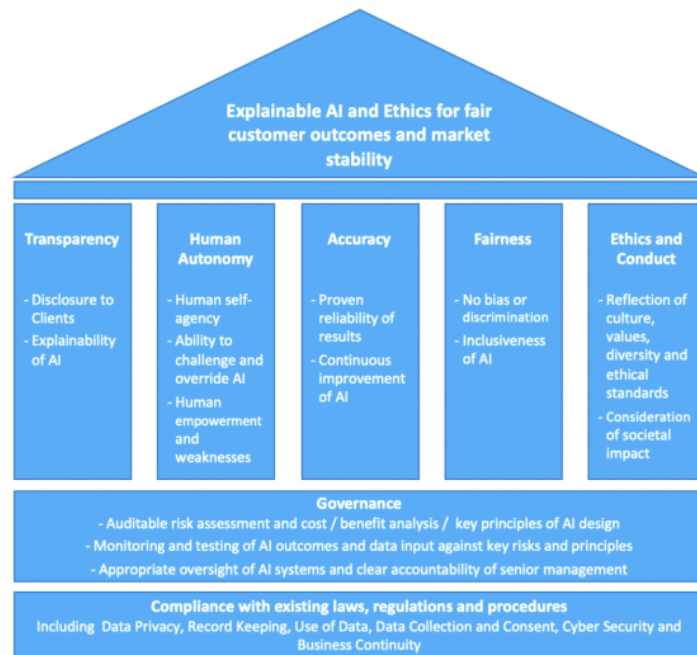


Figure 5 Accountability and Explainability Framework (Adams & Hagraas, 2020).

However, other key principles that are laid out are both the education and upskilling of the people who are responsible for the implementation and monitoring of the models, so they can properly understand the models. Additionally, continue with the current delegated authority regulations. This is something that is common practice within regulated industries, where people within the organisation have different levels of responsibility depending on their seniority. Based on personal experience with this method of governance, an example of how it works is illustrated in Table 3.

Level	Risk amounts
Senior Associate	£0 - £50,000
Manager	£50,001 - £250,000
Senior Manager	£250,001 - £1,000,000
Director	£1,000,001 - £5,000,000
VP	£5,000,001 - £10,000,000
Board member	>£10,000,000

Table 2 Example of Delegated Authority for authors' experience at Capital One

Within (Solanke & Liang, 2024), the authors go into detail about what explainable AI is and why it is so important in regulated industries. It discusses how, generally, there are trade-offs between performance and explainability of a model. It reiterates that as models become more complex, improving their performance often comes at the expense of explainability. It goes on to state that the development of “XAI in regulated industries is promising” (Solanke & Liang, 2024), with the evolution of regulations and advancements in XAI improving the transparency and interpretability of the models this will open up lots of opportunities within regulated industries, with examples given for Healthcare, Law and the target of this investigation Finance where it highlights the benefits for “credit scoring, fraud detection, and algorithmic trading systems” (Solanke & Liang, 2024) whilst demonstrating “decisions are made based on fair and unbiased criteria, ensuring compliance with laws” (Solanke & Liang, 2024).

2.1.2 Is Explainable AI the answer to Deep Learning in regulated industries

There are papers that make the point that explainable AI is only part of the solution to integrating Deep Learning into regulated industries. Some of these papers argue that even with explainable AI, this will not provide the transparency and visibility required by regulators. (Buckley, Zetzsche, Arner, & Tang, 2021) suggest that solely relying on explainable AI techniques is not enough. This is a broader business/industry issue that requires not only a technical solution, such as explainable AI, but also adding “humans into the loop”, making people in the business responsible for the actions and decisions made using the AI technology. This aligns with the sentiment of (Adams & Hagras, 2020) who highlight the option of delegated authority.

Additionally, other papers went a step further and argued that utilising explainable AI is an overcomplication of the solution and that the utilisation of traditional “interpretable” methods is more favourable due to the various possible complications that can arise from this lack of interpretability (Rudin, 2019) came to the conclusion that across many usecases that utilise data that can be prepared, not raw like pixels, that well refined parameters can have a larger impact on results than the variations between algorithms. Additionally, referring to results from (Razavian et al., 2015), which states that their regression model had similar, if not better, outcomes than “carefully tuned random forests, gradient-boosted decision trees, and neural networks” (Razavian, et al., 2015) however, they acknowledged that this was out of the scope of their study results and parameters utilised for this are not provided.

2.2 Deep Learning vs Traditional Methods

Numerous studies have examined the benefits of utilising Deep Learning techniques over more traditional methods. Identifying the core benefits of these more complex models, including (Anghel & Lupu, 2024), which deem the use of Deep Learning models as essential for banks and regulators to improve the understanding of future financial phenomena. While not using a financial services dataset (Korotcov, Tkachenko, Russo, & Ekins, 2017), the authors compared a variety of models on a medical dataset, which is another regulated industry in medicine. This found that the Deep Neural Networks outperformed algorithms like Random Forest, Linear Logistic Regression with Regularisation and AdaBoost Decision Tree. The results of this study illustrated that the Deep Neural Networks outperformed all the other test techniques across multiple metrics, and the Deep Neural Network with 5 hidden layers was ranked 1st out of the 9 models, outperforming the traditional methods, such as the Logistic Linear Regression or Random Forest, which ranked 6th and 7th, respectively. The full table is listed in Appendix D.

model	AUC	F1-score	ACC	Cohen-Kappa	Matthews	precision	recall	mean	rank
DNN-5	0.798	0.688	0.874	0.382	0.387	0.683	0.700	0.6448	1
LLR	0.809	0.604	0.789	0.286	0.316	0.584	0.746	0.5907	6
RF	0.760	0.612	0.795	0.277	0.297	0.594	0.700	0.5765	7

Table 3 Ranked Normalised Scores for Each Machine Learning Algorithm by Metric (Average over Eight Datasets for the Test Dataset) (Korotcov, Tkachenko, Russo, & Ekins, 2017)

Abbreviations: (DNN-5) Deep Neural Network with 5 hidden layers, (LLR) Logistic linear regression, (RF) Random Forest (Korotcov, Tkachenko, Russo, & Ekins, 2017)

As stated in other research and from the author's personal experience within the financial services industry. Regression and Tree-Based models are still the most commonly used models in financial services, mostly due to their ease of implementation and interpretability. The survey results from the Bank of England & Financial Conduct Authority (2019) found that over 50% of financial institutions used Tree-based models. It is worth noting that this survey does not distinguish between the various models used, as Natural Language Processing was the second highest surveyed outcome, but it is not commonly used for direct decision making. With over 80% of the models being used on structured data.

As part of this investigation, and pointed out by other researchers, this is believed to be a missed opportunity within not only the financial services industry but also wider regulated industries. This can be seen in the overall improved performance typically observed in structured data from Deep Learning Algorithms compared to both

decision tree and regression models. As illustrated in (Sirignano & Cont, 2018), they found that a Universal Deep Neural Network outperformed stock-specific Linear Models across roughly 500 stocks by approximately 10%, as shown in Figure 6. This could have a significant impact on the accuracy of financial decisions, affecting both company profits and the broader economy.

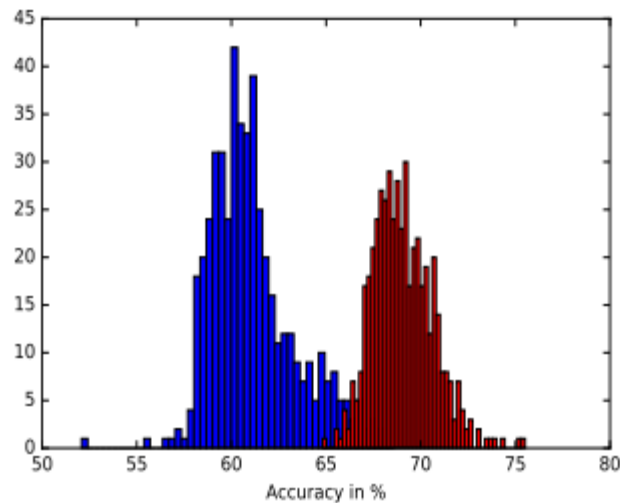


Figure 6 Comparison accuracy of a universal deep neural network (Red) with stock-specific linear models (Blue) (Sirignano & Cont, 2018).

This investigation looks to build on these results and benefits and many of the papers included above have either focused on why the explainability of models is important or the performance of various models this investigation looks to not only show that Deep Learning technique have improved performance over models that are currently in use but taking it a step further by combining methods of explain these Deep Learning model thus showing both why financial institution should use them but also how they can be implemented to adhere to the regulations that are currently in place and likely to expand in the future.

2.3 Explainable Deep Learning Methods

As discussed in Chapter 2.1.1, numerous papers have explored ways to explain Deep Learning Models and resolve the issue of these models being perceived as a “black box.” These methods are collectively referred to as Explainable Artificial Intelligence (XAI). Explainable AI methods are split into two types: Global and Local explainable methods. Most studies are consistent with the explanation of these two method types, with (Hakkoum, Idri, & Abnane, 2024) describing them as Global methods that provide a general explanation of the importance of a particular feature. While Local methods

explain how a particular input feature impacted a particular output point. Within regulated industries, both these methods are important for different reasons. Global methods are imperative for the Model Risk Management process, that is a set of frameworks by regulators that require financial services companies that utilise models to be able to effectively report on the effectiveness of the model and the risks associated with the use of the model, regardless of the complexity. As such, to comply with this regulation, Financial services companies will need to fully understand how the model they use works, including not only the data that goes into the model but also how features impact its outcome.

In contrast, Local explanation methods have a greater impact on the consumer, as they enable a company to examine an individual input and output, and identify the reasons the output was produced. This can build user trust, as it illustrates why a decision was made in their particular case. Additionally, regarding financial services, customers have the right to appeal a credit decision or request an explanation for the decision made. This is usually completed either by a set of rules, often referred to as “Hard Cuts,” which are naturally explainable. For example, a credit card company will not issue a credit card to individuals with an income of less than £10,000 or to applicants under a certain age. Alternatively, if an application passes the “Hard Cuts”, it will be subject to a model that requires the use of explainability methods, as these decisions are based on multiple variables.

Additionally, there are Model-specific XAI methods and model Independent methods. With the definitions and types of Explainable AI methods outlined above for the focus of this investigation, will be on model independent/ agnostic methods as these methods focus on “black box” techniques and do not rely on the structure of a specific model. As part of (Chen & Xiao, 2023) review of Interpretable ML they laid out the above types in a taxonomy shown in Figure 7. Additionally, Figure 8 provides a visual representation of the various techniques that will be part of this investigation, along with their alignment regarding model dependency and scope of interpretability.

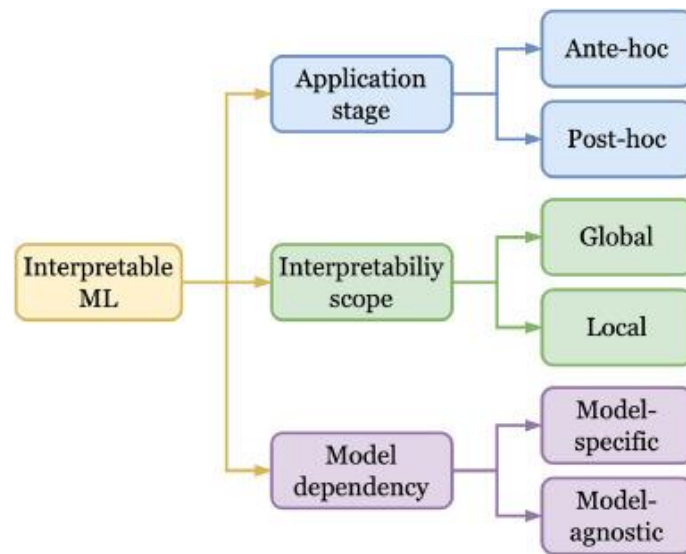


Figure 7 Taxonomy of interpretable ML techniques (Chen & Xiao, 2023)

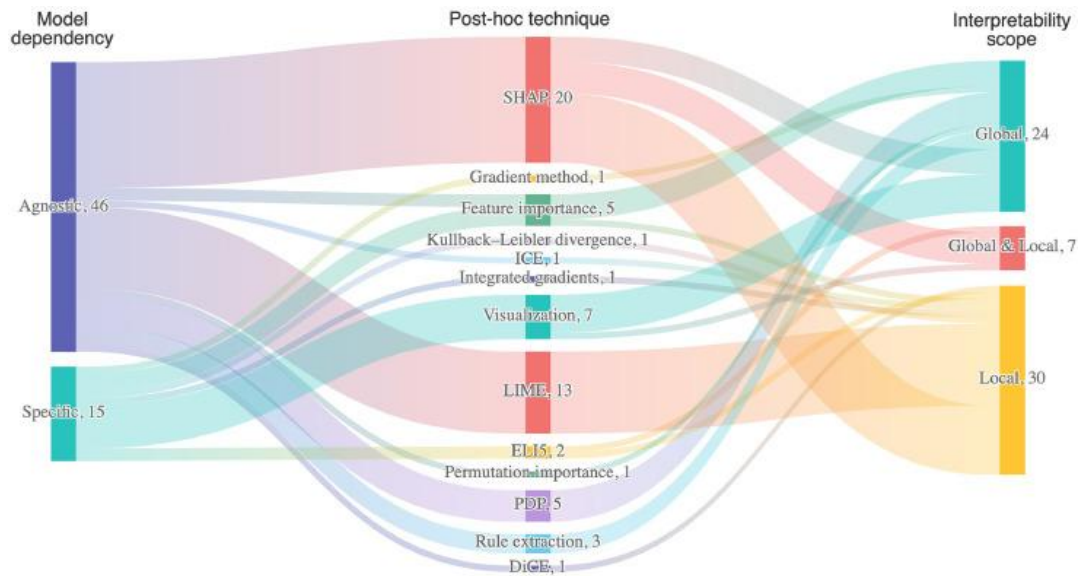


Figure 8 Sankey diagram depicting different model dependencies, post-hoc techniques, and interpretability scopes. (Chen & Xiao, 2023)

The 2 most commonly used post-hoc techniques from these studies were SHAP (Shapley Values) and LIME (Local Interpretable Model-Agnostic Explanations), with SHAP being able to be utilised for both Global and Local explanations, while LIME is used for Local only.

2.3.1 SHAP (Shapley Values)

Shapley values were originally induced as a game theory concept by Lloyd Shapley in 1953 with it not being linked to being used for explainable AI until much later by (Štrumbelj & Kononenko, 2011). However, it wasn't until (Lundberg & Lee, 2017) proposed new ways to estimate Shapley values and combined them with other explanation methods like LIME that it became more widely used.

It uses a game theory approach of estimating how each player contributed to the outcome. This is translated to XAI by changes made by players for features, assigning an importance value (SHAP value) to each feature for its contribution to the model output that it aims to explain. SHAP utilises an exploratory model, as shown in Equation 1. To evaluate the contributions of each input feature to the model.

$$EM = \varphi_0 + \sum_{i=1}^n \varphi_i t_i$$

$$\varphi_i(ML, x) = \sum_{t \subseteq x} \frac{|t|! (n - |t| - 1)!}{n!} [ML(t) - ML(t \setminus i)]$$

Equation 1 explanatory model (EM) equations (Lamane, et al., 2025)

“where n is the number of input variables, t_i is the abbreviation for the input variable (i). represents the model output with all simplified inputs toggled off, $\varphi_i \in R$ is the contribution of the variable i to the ML model, t is the subset of the features used in the model that does not include the feature value i , $ML()$ is the original prediction model, and $\setminus$ is the difference set notation for set operations.” (Lamane, et al., 2025).

2.3.2 LIME (Local Interpretable Model-Agnostic Explanations)

LIME is another popular method for explaining black box algorithms. In contrast to SHAP, LIME focuses on local explainability when not used in conjunction with other methods, as mentioned in chapter 2.3.1. LIME was originally proposed by (Ribeiro, Singh, & Guestrin, 2016) as part of their paper to show how you can explain classifiers. LIME works by focusing on an instance of the model and adjusting the feature values, then using the original model to find how that would have impacted the model outcome. As shown in Figure 10. Which is an example of a 2 feature complex none linear model. However, LIME works by focusing on a local set of values to produce a linear model (the dashed line). Focusing on the bold red cross, LIME would adjust the feature values to assess their impact on the model's output. In this example, a negative change to Feature 2 is unlikely to alter the final output, whereas a small positive change would likely shift the outcome from pink to blue. However, for Feature 1, it would require a significant change in either direction to impact the model's

output. Lime can be formally described by Equation 2. Where an explanation is defined as “model $g \in G$, where G is a class of potentially interpretable models..... g acts over absence/presence of the interpretable components $\Omega(g)$ be a measure of complexity and $L(f, g, \pi_x)$ be a measure of how unfaithful g is in approximating f in the locality defined by π_x ” (Ribeiro, Singh, & Guestrin, 2016)

$$\xi(x) = \underset{g \in G}{\operatorname{argmin}} L(f, g, \pi_x) + \Omega(g)$$

Equation 2 The equations for deriving the explanation produced by LIME (Ribeiro, Singh, & Guestrin, 2016)

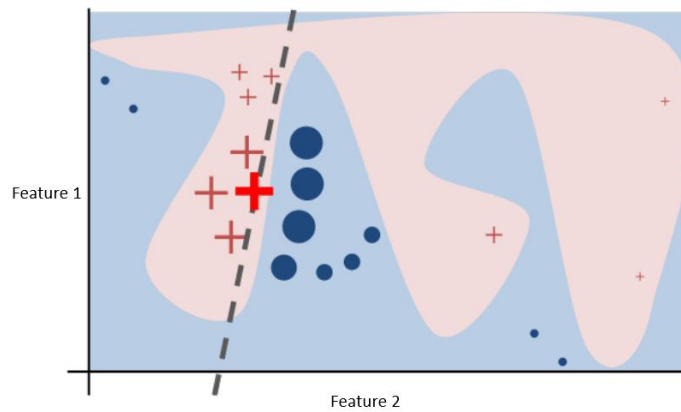


Figure 9 A Toy example of to present intuition for LIME. The bold red cross is the focus instance that is being explained. With the dashed line is the learned explanation for this local instance and the pink and blue is the unknown black-box model's decision functions (Ribeiro, Singh, & Guestrin, 2016)

As discussed throughout this literature review, there are papers that have covered the benefits of using Deep Learning algorithms over traditional algorithms, and there are studies that have highlighted issues with why Deep Learning algorithms may not be possible to use, especially due to the limitations of explainability, and lastly, there are academic papers that focus on techniques that can be utilised to resolve this lack of explainability. However, the goal of this investigation is to make a significant contribution to the field by providing practical examples of how these elements can be combined to enable regulated industries to fully benefit from the use of Deep Learning while maintaining regulatory compliance.

Chapter 3

Design and Implementation of the Experiment

3.1 Hardware and Software

The hardware and software that have been utilised for this investigation were Python as a programming language and utilising Jupyter Notebook as an Integrated Development Environment (IDE). This was run on a personal PC with specs: Processor - Intel(R) Core(TM) Ultra 5 245KF 4.2 Ghz, 32GB RAM, GPU - NVIDIA GeForce RTX 4060 on OS - Windows 11, version 24H2. Due to the use of this hardware, certain parameters had limitations based on the tools available for this investigation. These parameters and decisions are discussed throughout chapter 3.

3.2 Data preparation

Before building a model, it's essential to organise the data and ensure its quality meets the required standard. The performance of both Deep Learning and traditional models heavily relies on the quality of the input data. Additionally, what data is used and how it is used are crucial to ensuring the explainability, trustworthiness and transparency of the model and how it complies with regulations.

Data preparation steps include data profiling, which ensures the data is appropriate for the model's task (Abdallah, Du, & Webb, 2017). This investigation uses a widely utilised financial services dataset that is representative of what banks access when building a credit risk model. Such models help lenders decide on credit applications for mortgages, loans and credit cards with varying criteria across lenders. The dataset focuses on Lending Club's loan data (LendingClub, 2019) The target used for this investigation is whether a customer paid off their loan or "charged off", with the goal of predicting an applicant's likelihood of repayment.

The decisions and processes implemented for this investigation are based upon best practices and would be followed regardless of model type.

The first step in preprocessing was to remove rows that did not have completed loans, defined as either "Fully Paid" or "Charged Off", these were updated to be 1 and 0,

respectively. Other statuses, indicating ongoing loans or those outside of credit policy, were excluded because their outcomes were either unknown or because they should not be included as standard accounts.

The next stage focused on excluding features other than the target variable that are only available after loan acceptance, such as settlement data. A complete list of exclusions is provided in the code and Appendix A. Next, rows with missing data were examined. The approach differed depending on the amount and type of missing data, as well as real life implications of NULL values. These decisions were as follows:

- Features with high levels of NULL values were deleted as they provided no distinguishing ability for the model. A complete list is provided in the code and Appendix A.
- The remaining features were split into categorical and numerical data types:
 - For the categorical features, it was not possible to determine values with certainty. Although methods exist to approximate values, such as in (Kang, 2013). However, it did not seem valid to generate a missing data such as postcode for an application that may be incorrect and thus go against the trust and transparency we are aiming to achieve. As such, these specific rows were deleted. The features involved are listed in Appendix A.
 - In contrast, the numerical features, it was reasonable to believe that the corresponding value would be 0, with reasons given for each feature involved listed in Appendix A.
 - Next, variables that are either redundant, are free text fields and data that is not allowed for the purpose of this model, such as address information, while address data can be used in fraud checks as part of an application, it cannot be used for a creditworthiness decision, have been excluded with a list in Appendix A.
 - After removal of all unusable data, we can now convert it into formats that the model can use. This includes mapping categorical variables to clear numeric equivalents, as seen in Appendix A.

The training and datasets were prepared, categorical features encoded using `sklearn.OrdinalEncoder` (scikit-learn, 2025). The data was then randomly split 67% training and 33% test sets. `MinMaxScaler` was used to normalise the data, ensuring features contribute equally during training (scikit-learn, 2025).

This process was repeated for a sub-sample of the data that has fewer features for the purpose of model performance comparison.

3.3 Models

The primary model to be implemented for demonstrating explainability and performance is a Deep Neural Network. While the focus of this investigation is not to build a cutting-edge model, it aims to demonstrate its overall capabilities and the

explainability of this model through various means, enabling it to be used in regulated industries. Additionally to this, there are restrictions on the size and scale of the model that can be built, both with the time constraint of this investigation, and with the equipment available, as cutting-edge deep learning models can cost in the hundreds of thousands of pounds to build (FP Data Solutions Team, 2024).

The model was built using the TensorFlow model package, following a structure that defines the size and shape of the neural network. For the purpose of this investigation, the model has an input layer with one node per input feature, 10 hidden layers, which exceeds the typically defined 3 hidden layers of Deep Learning, as mentioned in Chapter 1.1.1, with 50 nodes each, and an output layer with 2 output options. For other parameters, the decision was taken to use commonly used parameters that are the most representative of the general utilisation of the model. A summary of this is in Table 5. A dropout rate was randomly selected between 0.2 (Li & Ke, 2024) and 0.5 (Pauls & Yoder, 2018) for each hidden layer, as this range was based on various studies aimed at preventing model overfitting. Next, the initial learning rate was set to 0.005, which differs from the Adam optimiser's default of 0.001, to be more in line with other common learning rates. (Verma, Meenpal, & Acharya, 2018) There are pros and cons to higher and lower learning rates. A lower rate is slower to train but can reach a more optimal solution, whereas a higher rate can also become stuck at suboptimal solutions. Next, a batch size of 16 is used due to the small size of the dataset in Deep Learning terms, and an epoch of 10 is used. These values determine the size of each data sample in the entire dataset and the number of iterations of the entire dataset on which the model is trained. These values were heavily influenced by the hardware available for this investigation, as larger numbers generally require more memory. The Adam (Adaptive Moment Estimation) was selected for its widespread use and the efficiency of the model. ReLU (Rectified Linear Unit) activation function was used for the hidden layers for its ability to handle complex nonlinear relationships, as are common in this dataset. The sigmoid activation was utilised for the output layer due to its probabilistic output being ideal for binary classification models.

Parameter	Value
Number of hidden layers	10
Nodes per layer	50
Dropout rate range	0.2-0.5
Initial learning rate	0.005
Epochs	10
Batch size	16
Optimizer	Adam
Loss function	Binary cross-entropy
Hidden layer activation	relu

Output layer activation	sigmoid
-------------------------	---------

Table 4 Table of Neural Network parameters

3.4 Explainability methods

While the explainability of the model output is discussed throughout Chapter 2. It is not the only part of ensuring a model is explainable, transparent, and meets regulatory requirements. As illustrated in Chapters 3.2 and 3.3, clear explanations have been provided to demonstrate how data has been utilised and decisions made regarding the data itself and the model's configuration. This leaves the model output to be explained. For this, the 2 most commonly used XAI techniques have been utilised, these being LIME and SHAP, as shown in Figure 8. As discussed in Chapter 2.3 both methods are model agnostic, and by utilising both methods, this investigation can give examples of both global and local explainability, which is ideal for the use case of this investigation, as it aims to give a practical example that can be utilised not just across models but across industries.

Regarding LIME as a local explainer, there are several ways to illustrate the data. An example of which is shown in Figure 10. The example illustrates the probability of each outcome, as shown on the left. The impact each variable has on the outcome is displayed in the middle, along with an explainable reason and the actual feature values on the right. This method was chosen because it meets the requirement of transparency, showing not only the data used in making a decision but also the specific impact that each feature value had on the result. In this investigation, the output could be communicated back to the applicant, allowing them to address any concerns and thereby build trust within society regarding the decision making process. Furthermore, the model output regarding the probabilities of each outcome, where probabilities were close, could be used to flag a manual review and in doing so, address concerns described in Chapter 2.1.2.

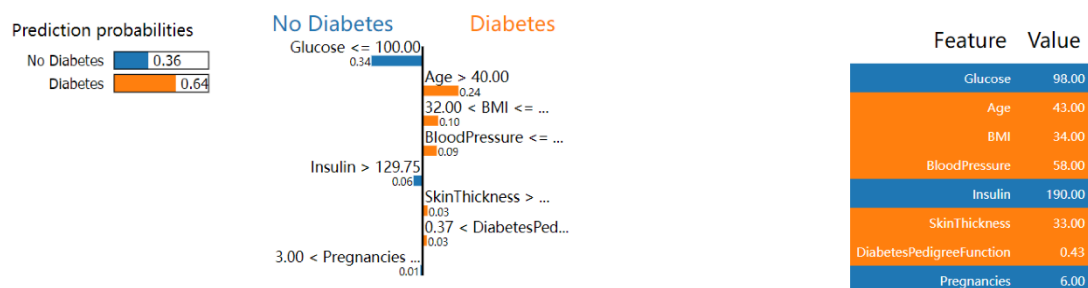


Figure 10 Example of LIME output plotted (An, Zhang, & Joe, 2023)

SHAP, however, can be used for both global and local explanations when used in conjunction with LIME as discussed in Chapter 2.3. This is commonly done through

various methods and plots. Examples of plots used to explain the model as a whole are the Force plot, mean SHAP plot, Beeswarm plot and Dependence plot. Example a) of Figure 11. This is an example of a Beeswarm plot that can be used to explain the impact each feature has on the final result, with the order of features being descending in impact, but additionally, you can see if a large or small value for this feature has a positive or negative effect on the output value. However it cannot give individual reasons for specific instances, this is where SHAP combined with LIME as develop by (Lundberg & Lee, 2017) can be used to explain individual outcomes and how they compare to the overall result, providing both Global and Local explanations and was the only model agnostic method with the facilities to do this of the 61 studies investigated by (Chen & Xiao, 2023).

These methods were chosen for their versatility in explaining all aspects of the models and applicability across various industries, aligning with the aims of this investigation.

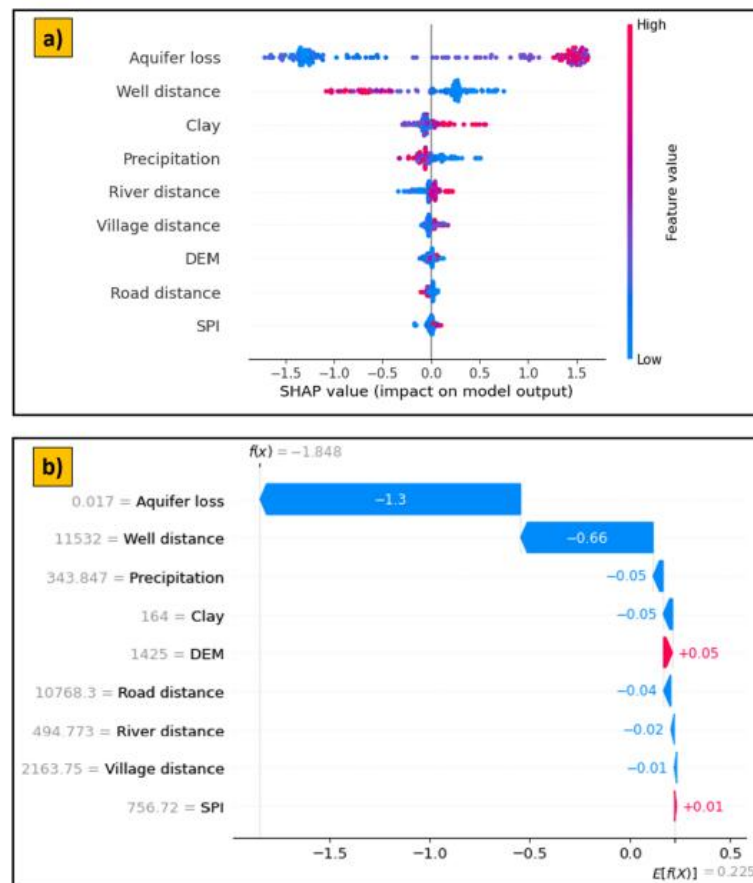


Figure 11 A) SHAP plot illustrating global explanation and B) waterfall plot showing Local explanation (Mohamadifar, Gholami, & Golzari, 2022)

Chapter 4

Results

4.1 Pre-processing Outcomes and Data Structure

Both deep learning and traditional models (Nyandwi , 2020). When directly compared, they were trained and tested on the same data. The outcomes of this preprocessing process are shown below in Table 5, for both the main and subset datasets, to enable performance comparisons across different data sizes

- Main: 145 Features & 2,260,668 Instances → 84 Features & 1,207,019 Instances
- Subset: 27 Features & 396,030 Instances → 22 Features & 371,384 Instances

Due to the imbalance of the target variable of the selected dataset, approximately 80% of accounts Fully Paid, and 20% Charged Off. Selecting an appropriate performance metric is crucial. As with this dataset, a model predicting Fully Paid all the time would achieve an 80% accuracy but wouldn't identify any Charge Off accounts. Identifying accounts that are likely to charge off is critical, as mentioned in Chapter 1, charge offs significantly affect lenders' profitability. Different lenders have varying charge off rates based on their lending types and demographics. For instance, subprime lenders anticipate higher charge off rates due to targeting riskier applicants but compensate with higher interest rates.

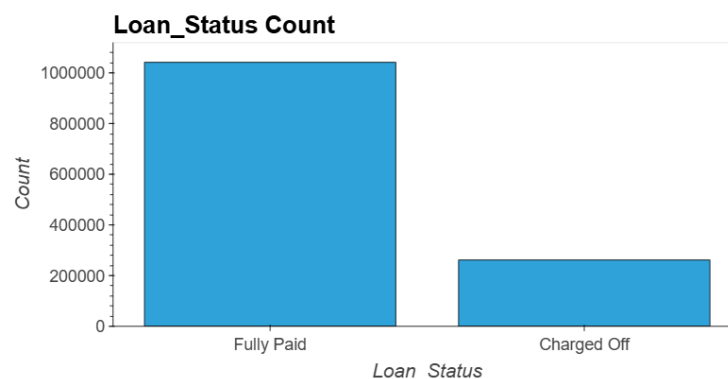


Figure 12 Target variable distribution

4.2 Model Performance

For completeness of comparing Deep learning against traditional models, the models were trained and tested against the main dataset and a smaller subset.

As stated in Chapter 4.1, the selection of performance metrics is vital when comparing any models and models can and should be optimised to target key insights and outcomes. In the case of this investigation, this was correctly classifying whether an applicant would likely fully repay their loan or charge off. As lenders want to maximise accounts that will make repayments and minimise those that will charge off to maximise profits. As such, the model comparisons will be focused on:

- ROC Score, as it focuses on the model's ability to distinguish between 2 classes, focusing on True Positive Rate and False Positive Rate
- Precision, which focuses on identifying True Positives (TP), and is calculated as follows $\frac{TP}{TP+FP}$. This is crucial because False Positives (FP) are accounts that will likely charge off and result in financial losses.
- Specificity, the focus of this metric is to identify True Negatives (TN) and is calculated as follows $\frac{TN}{TN+FP}$. This metric focuses on the ability to accurately identify the negative instances, similar to Precision False Positives (FP), which can result in financial losses.
- F1-Score, which looks to produce a harmonic mean between Precision and Recall (this can be Specificity for the negative instance and Sensitivity for the positive instance), providing a single measure of model performance that is especially effective for datasets with large imbalances, like in this instance. This is calculated as $F1 = \frac{2*TP}{2*TP+FP+FN}$.

While training the Deep Neural Networks on both the Main and subset datasets, the AUC and Validation AUC (VAL_AUC) scores of both models reached plateaus, indicating that the models are no longer improving, as shown in Figures 13 and 14.

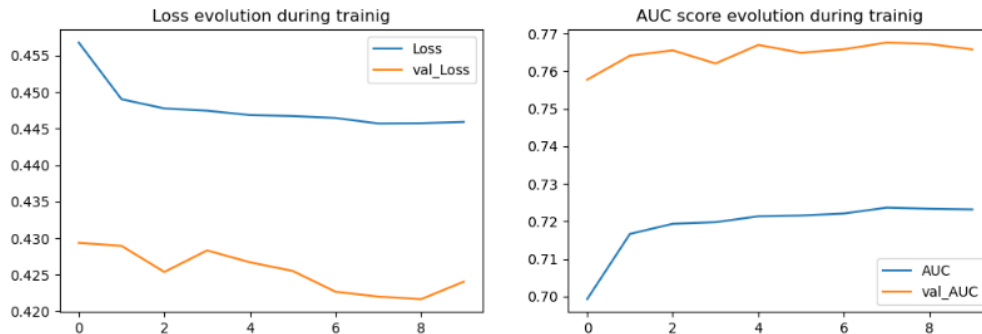


Figure 13 Learning evolution graphs for DNN

Investigation into how to Increase Deep Learning Utilisation in Regulated Industries

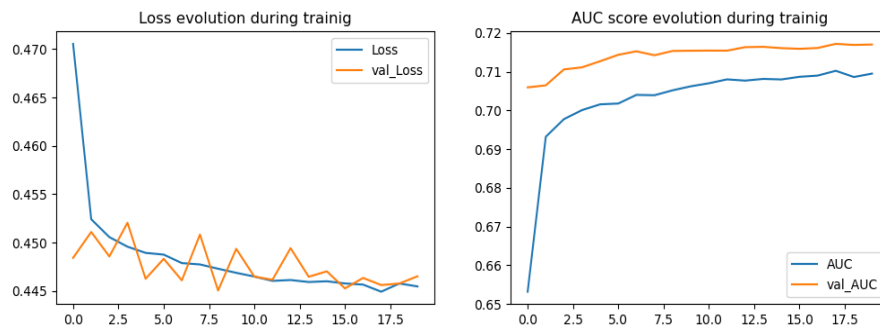


Figure 14 Learning evolution graphs for DNN on the subset dataset

This is commonly caused when the model has become trapped in a suboptimal solution. While this can sometimes be resolved by adjusting hyperparameters, this investigation is not focused on hyper-optimisation of Deep Learning models. However, even without the optimisation of the deep learning models' parameters across both the main and subset datasets, the Deep Learning Neural Network outperformed both the Traditional Models (Logistic Regression and Random Forest models) in ROC scores, shown in Figure 15 and Table 5. With only the Deep Neural Network being considered acceptable in accordance with the definition from (Mandrekar , 2010)

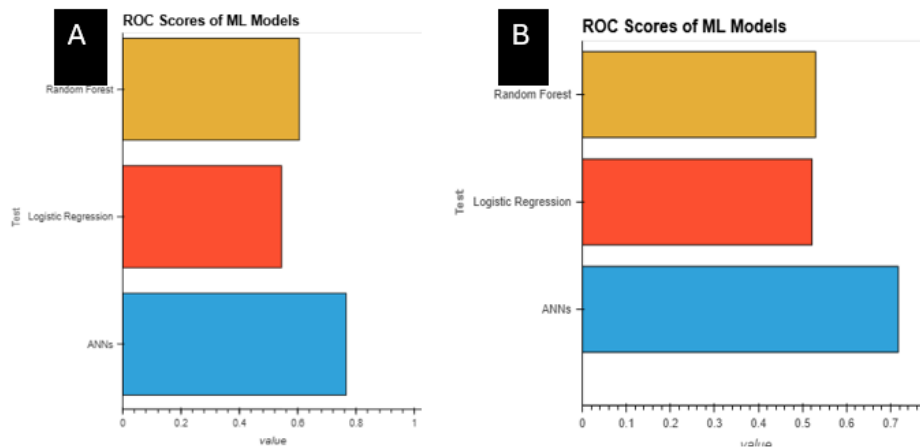


Figure 15 Model ROC Score comparison A) on full dataset, B) on subset dataset

Model	Main	Subset
Deep Neural Network (ANN)	0.766	0.718
Random Forest	0.605	0.530
Logistic Regression	0.544	0.521

Table 5 Model ROC Score comparison

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Based upon this initial metric, the Deep Neural Network outperforms both traditional models in correctly identifying positive cases and minimising false positives. However, this is not the only metric that was investigated.

Next, this investigation will compare the Precision, Specificity and F1-Score, of the 3 models. When looking at the results shown in Figure 16.

- When comparing the basic accuracy across the 3 models, they are very similar, with less than 0.1% difference between them all. This gives the perception that there is no performance difference between the models.
- However, when the focus is changed to the target metrics, Precision, Specificity, which is the Recall of the negative class, and F1-Score.
 - We can see that the Deep Neural Network outperformed both traditional models on the weighted average of both the Precision and F1-Score and produced similar results for Recall.
 - Additionally, when looking at Specificity, which is the Recall of the 0 column, the Deep Neural Network was 140% the performance of the Random Forest and 190% the performance of the Logistic Regression Model

<pre>Test Result: ===== Accuracy Score: 80.89% CLASSIFICATION REPORT: 0.0 1.0 accuracy macro avg weighted avg precision 0.523172 0.819972 0.808946 0.671572 0.762756 recall 0.100821 0.978055 0.808946 0.539438 0.808946 f1-score 0.169062 0.892064 0.808946 0.530563 0.752687 support 23626.000000 98931.000000 0.808946 122557.000000 122557.000000 Confusion Matrix: [[2382 21244] [2171 96760]]</pre>	Deep Neural Network
<pre>Test Result: ===== Accuracy Score: 80.85% CLASSIFICATION REPORT: 0.0 1.0 accuracy macro avg weighted avg precision 0.523605 0.816271 0.808481 0.669938 0.759852 recall 0.072293 0.984292 0.808481 0.528293 0.808481 f1-score 0.127046 0.892442 0.808481 0.509744 0.744892 support 23626.000000 98931.000000 0.808481 122557.000000 122557.000000 Confusion Matrix: [[1708 21918] [1554 97377]]</pre>	Random Forest
<pre>Test Result: ===== Accuracy Score: 80.91% CLASSIFICATION REPORT: 0.0 1.0 accuracy macro avg weighted avg precision 0.550262 0.814025 0.809101 0.682144 0.763178 recall 0.053289 0.989599 0.809101 0.521444 0.809101 f1-score 0.097168 0.893266 0.809101 0.495217 0.739798 support 23626.000000 98931.000000 0.809101 122557.000000 122557.000000 Confusion Matrix: [[1259 22367] [1029 97902]]</pre>	Logistic Regression Subset

Figure 16 Detailed Classification report across the Deep Neural Network, Random Forest and Logistic Regression Model.

Based on the metrics defined in this section, it is clear that Deep Learning models have the potential to outperform traditional models, with the Deep Learning model identifying application outcomes more accurately than the traditional models. This was completed without additional optimisation of the Deep Learning model. It is important to note that model performance can vary depending on the data type and use case. This is intended to be a representative example of the majority, not a definitive rule.

4.3 Model Explainability

To achieve the goal of Objective 2, this investigation looks to prove that complex Deep Learning models can be explained in a way that would meet the regulators' requirement of explainability, trustworthiness and transparency.

This investigation has utilised both LIME and SHAP packages, as described in Chapter 3.4. Initially, this investigation will demonstrate how these tools can be used to explain the outcome of a particular instance, in this case, an application. This model would be used to determine whether an application is accepted or not. As such, to demonstrate that a model can be trusted and a decision can be explained to an applicant, this investigation initially utilised the LIME package, as illustrated in Figures 17 and 18.

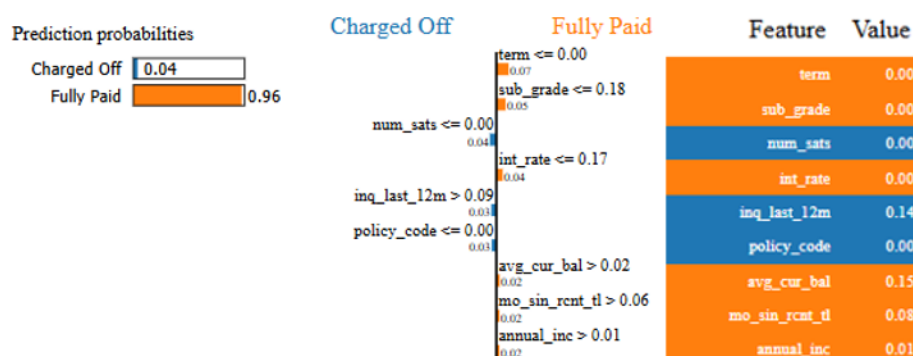


Figure 17 LIME explainer model Column Stack output Fully paid example

This output, as discussed in Chapter 3.4, can be used to:

- Inform an applicant, an auditor, or the company why a particular application, in this case, was predicted to be fully paid and should be accepted.
- Illustrates that the term length, sub-grade and interest rate of this applicant are positive factors. This could indicate that loans with lower interest rates and shorter terms are more likely to be repaid.
- Additionally, inform that the number of satisfactory accounts (num_sats), the Number of credit inquiries in the past 12 months (inq_last_12m) and the public

code are all negatives for this application. As this would indicate a limited account history known as “Thin”, which can be more risky, and they have made multiple credit enquiries within 12 months, which can normally indicate credit desperation.

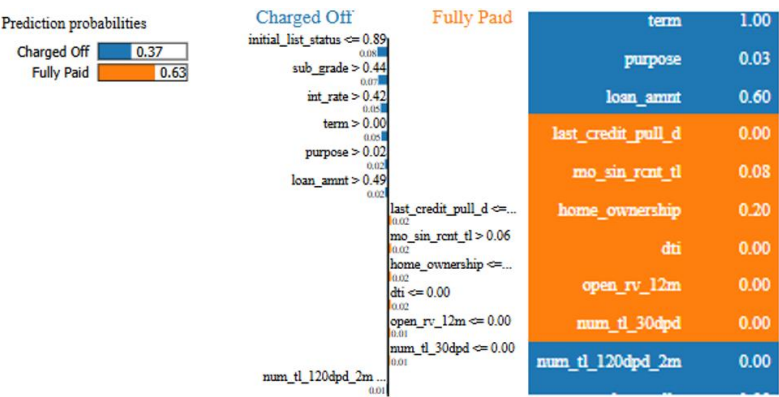


Figure 18 LIME explainer model Column Stack output Fully paid example

Similar to the example given in Figure 17, Figure 18 has a list of reasons that impact whether an applicant is likely to repay the loan. However, as you can see in the prediction probabilities on the left, it gives a value for which outcome is likely. This has the potential to be used as a manual review mechanism, where if these values are within a certain range, they are reviewed by a human. A glossary has been provided for column abbreviations in Appendix B.

Additionally, by combining these numbers for all instances, a guide to the most impactful features can be produced, as shown in Figure 19. This shows that how long ago the applicant had a credit pull, term length, subgrade of the loan and interest rate are the biggest contributors to whether a loan will be repaid. While Figures 17 & 18 are good for giving feedback to individual applicants, the data in Figure 19 could be used to influence credit policy, as it could indicate that a particular combination of loan length, interest rate and grade is unlikely to be repaid, making it an unprofitable product. However, this is not a substitute for Global explainability achieved with SHAP.

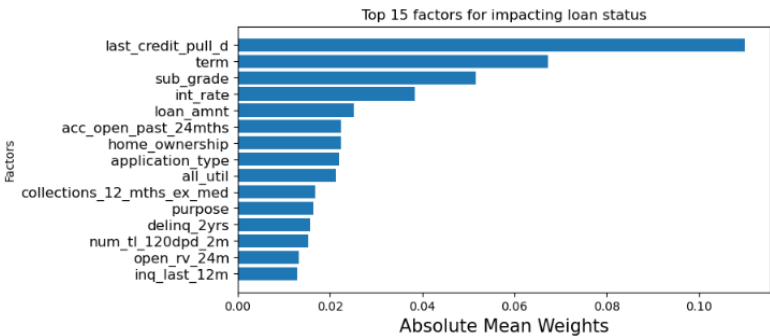


Figure 19 Average absolute LIME weightings to produce a ranking for the most impactful features

This investigation then built upon this explanation of specific instances and aimed to utilise SHAP to provide a comprehensive explanation of the model. This is illustrated below with the beeswarm chart in Figure 20, where each dot represents an instance. The colour of the dot indicates its value, with the bluer it is, the lower the value, and the redder it is, the higher the value. The distance from 0 reflects the impact it has on the model. As can be seen, there are similarities between the estimation based on the LIME absolute Mean weights and the beeswarm chart, both giving an indicative order of the impact a feature has, while the SHAP beeswarm chart allows the user to see how the value impacts the model output, which cannot be achieved with LIME on its own. Examples from Figure 20. It appears that shorter term lengths, as shown in blue, have a positive impact on loan outcomes, while longer term lengths, as shown in red, have a negative impact. This is clear for binary options, but holds true in the example of interest rates, which is a continuous metric, as the interest rate decreases, going from blue to purple to red, the impact becomes more negative. This is invaluable when selecting new products and explaining decisions, such as affordability, to auditors and regulators, as lenders may no longer want to offer long-term products if customers are unlikely to make repayments.

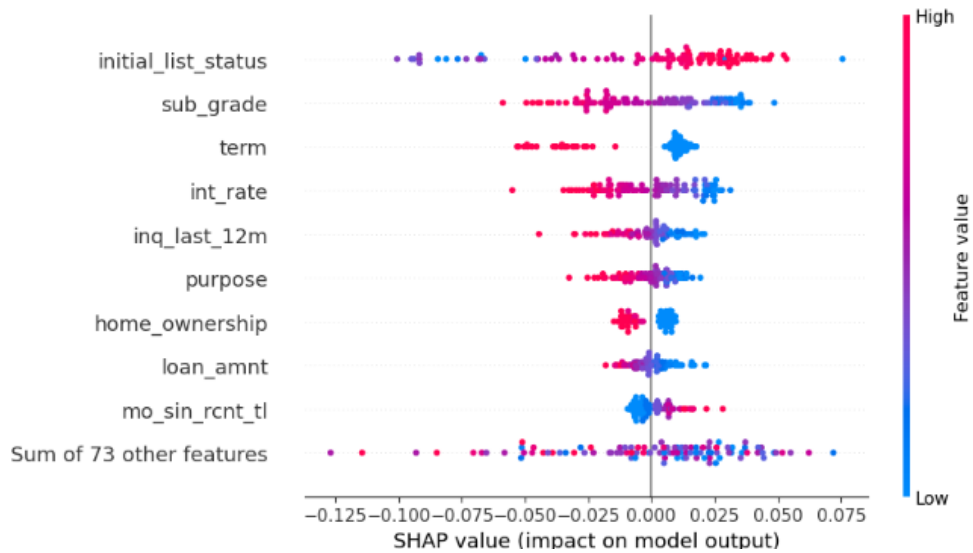


Figure 20 Beeswarm plot of SHAP values

This is where using SHAP with LIME can add an extra level of explainability, ultimately enabling this investigation to provide both global and local explanations, and combine them to demonstrate how a particular instance compares to the overall dataset. This has been achieved using a waterfall output in Figure 21 that shows $E[f(x)] = 0.807$, which is the average log odds of the model with $f(x) = \ln\left(\frac{P}{1-P}\right)$. To convert this into the original probability, you would use the equation, $P = \frac{e^x}{1+e^x}$. This would give an average probability of 0.691. With this particular instance, having $f(x) = 0.838$, giving a probability of 0.698, with the effect of each feature being shown with the size of the

bar and how it moves this instance output in relation to the mean output.

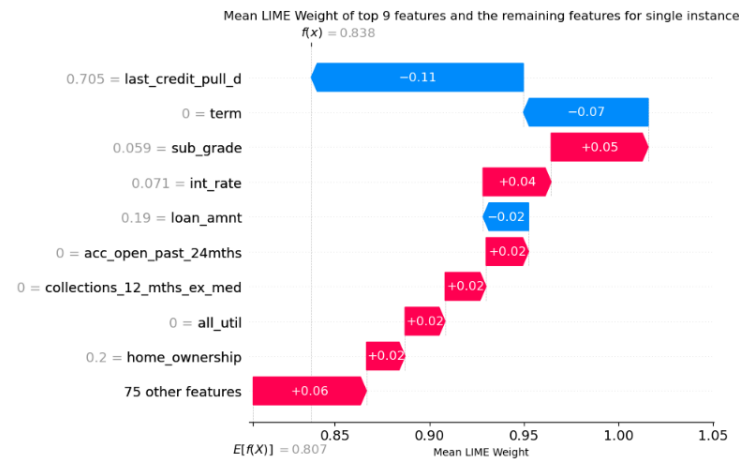


Figure 21 Waterfall chart of SHAP with LIME

Based on the balance of the data, it is expected that the average probability would be biased towards the Fully Paid. With each bar indicating how this particular applicant's features affect the prediction in relation to those of an average applicant. This can be used to produce products similar to those that compare credit scores based on various factors (Experian, 2025).

With these examples, it is the belief of this investigation that Objective 2 proving that Deep Learning models, "black box methods", are in fact explainable and can be presented so that the outcomes can adhere to the regulatory requirements of being explainable, transparent and trustworthy as the overall working of the model and how features impacted the outcome can be demonstrated to the general public to help them understand and improve trust in the rational, but also to auditors to show how and why data was used and the impact that data had on the businesses decision making.

Chapter 5

Discussion and Conclusion

5.1 Recommendations

Based upon the results and methods discussed above, the following list of recommendations has been produced:

- Recommendation 1 - Data used for training Deep Learning algorithms must comply with standard regulations regarding data usage and protected data, specific to the industry and use case.
- Recommendation 2 - Preprocessing steps should be clearly articulated, detailing why data was removed, transformed or substituted in a manner that can be explained to the general public, auditors and regulators.
- Recommendation 3 – The model selected and target outcomes should not aim to identify or disadvantage any protected characteristic and should be used for a purpose that is permitted by the industry regulator.
- Recommendation 4 - The selected model should include all parameters along with a justification and a summary of the intended impact of these parameters on the model. Although this information will not be shared with the general public or competitors, it facilitates auditing if required and enables the seamless handover of the models to other team members or teams within the business.
- Recommendation 5 - The outputs of the model should be explained at both the Global and Local levels, as demonstrated in this investigation. Although the exact techniques may vary, a model's overall performance, as well as the general impact, scale and direction of its features should be explainable. Additionally, it should be possible to explain individual instances that have been processed by the model, with reasoning presented in clear, simple language.
- Recommendation 6 - In addition to the previous recommendations, an expansion of the delegated authority process used by a number of industries should be implemented more widely. Depending on the application of the model, different levels of seniority should be involved in understanding and approving the model. Furthermore, others in the business should understand the model to perform manual reviews of fridge cases or borderline decisions

and to monitor the model's ongoing performance as part of effective model monitoring practices.

5.2 Discussion of and evaluation of Objectives

The goal of this investigation was to increase the uptake of Deep Learning methods within regulated industries. To achieve this goal, this investigation outlined 3 objectives in Chapter 1.3.

5.2.1 Objective 1 discussion and evaluation

Focused on comparing the performance of Deep Learning models with traditional models, the two traditional models that were selected for this investigation are the Random Forest and the Logistic Regression model. These models were utilised as representative models that are currently widely utilised within the financial services industry and other regulatory industries. Additionally, 2 sizes of datasets were used: the main and a subset of the main. As discussed in Chapter 4.2, the Deep Neural Network outperformed the traditional models across both dataset types, this was across the key metrics identified:

- ROC: The Deep Neural Network achieved results on average of 134% of the traditional models.
- Produced near equal results when looking at weighted average Precision, Recall and F1-Scores
- When focused on the identification of Charge-off accounts, the Deep Neural Network produced an average Specificity result of 164% and an F1-score result of 154% compared to that of the traditional models.

Now, we will cover the “perceived” extra time and effort required to build these models and then explain them. In the context of this investigation, the majority of the development time, once the methods were chosen and researched, was on the data preparation, thanks to the packages available in python the difficulty to implement the models was consistent across all models with additional parameters being required for the Deep Neural Network model itself and then the explainability packages required the model, data inputs and model outputs. Additional charts and graphs can be generated for all models with minimal code adaptation. Additionally, this investigation did not focus on hyperparameter optimisation and pushing the boundaries of Deep Learning, which could increase the build and training times. However, this investigation didn't focus on creating the best Deep Neural Network possible, but rather aimed to present a blueprint with practical examples that can be easily adapted and utilised across industries for various use cases, demonstrating that these techniques not only have the potential to improve performance but are also usable across regulated industries and make a significant contribution to the field.

Based on these results, the author believes that Objective 1 can be considered complete, and this investigation has provided evidence that Deep Learning techniques are worth pursuing by regulated industries due to their performance benefits.

5.2.2 Objective 2 discussion and evaluation

Objective 2 of this investigation was to enable the utilisation of Deep Learning techniques by demonstrating that these techniques aren't unexplainable and the "black box" element of the model can be explained in ways that adhere to regulations. While these regulations differ between industries, they generally fall within the 5 areas listed below: Safety, Security and Robustness; Appropriate Transparency and Explainability; Fairness; Accountability and Governance; Contestability and Redress (Gallo & Nair, 2024)

Throughout this investigation, steps have been taken to provide a blueprint that can be adapted by businesses to enable them to adhere to these 5 areas.

This investigation began with the sourcing of data. In this case, a publicly available dataset was chosen, as it is a replication of a real dataset that would be available to a financial services company. The data was then prepared to ensure that it was appropriate and did not discriminate against protected characteristics. It is worth noting that the usable data varies by use case and industry, as highlighted in this investigation. While financial services can use an address for fraud purposes, it cannot be used for creditworthiness modelling, as it can unfairly disadvantage people from poorer areas. These preprocessing steps are integral to the fairness aspect of the regulation, as well as to explainability and transparency, since the ability to explain and demonstrate the data used to make a decision is just as important as explaining the output. However, aspects of the process should be the same regardless of the model being used, as was the case for this investigation. This is also the same for the first area around safety, security and robustness as how the data that is used and is being stored is not only a part of AI regulations but data usage in general, this investigation does not go into detail around data security and safety but acknowledges it's importance to the area however it is outside the scope of this investigation.

This leaves us with the final part of the model build process, which utilised XAI techniques in this case, specifically SHAP and LIME, to explain how the model makes its decisions and the outcome of those decisions for individual instances, as well as the general impact on the whole model. This has been demonstrated throughout Chapter 4.3, where the features with the highest impact on the model were highlighted, and how each of the features impacts the outcome, whether in a positive or negative way. Additionally, when looking at local explainability, these methods enable us to explain the outcome and reasoning for a specific instance. In utilising these techniques, it elevates the "black box" terminology and enables models to be appropriately transparent and explainable. This doesn't mean that a business needs to give over proprietary methods and tools, but it must be able to explain how its decision making works. People will be able to contest decisions and have the reasons

for their specific case outcome. This will be provided with specific reasons that won't be based solely one factor as that is not how these decisions are generally made or be told a vague answer of based on "our complex algorithm", but rather these methods enable the user to say the top X reasons for this decision were in the case of financial services that the applicant has a CCJ, their debt to income ratio is over 0.3 and so on. This is something that an applicant can understand and look to improve, but also provides validity to the decision making that can be illustrated if an applicant appeals a decision or an auditor or regulator enquires.

However, these models are not perfect and unlikely to be in the near future, as such part of the recommendations outlined in Chapter 5.1 states that human intervention and knowledge will be required for the foreseeable future as is the case currently required as discussed in Chapter 2.1.2, the builders of the models are responsible for the decisions and outputs made from these models and as such have a burden of responsibility to understand how they work and what impacts the outcome. As suggested in the design of the experiment, having someone review outcomes that are borderline between classes is still believed to be a good idea and shows businesses are taking accountability to ensure decisions are made fairly, but with the results demonstrated, these cases should be less frequent with the improved classifications achieved with the Deep Learning model.

Based on the explanations that can be generated and the examples provided, the author believes that Objective 2 can be considered achieved, as all aspects of data curation, preparation, model selection, design, and outputs can be clearly explained.

5.2.3 Objective 3 discussion and evaluation

As outlined in Chapter 5.1, this investigation presents a set of recommendations that companies within regulated industries can utilise to comply with the current regulations established by their industry. These recommendations are not industry-specific and can be used as a guide in combination with the practical example illustrated in this investigation of a model within the financial services industry. These recommendations ensure that the key areas laid out in (Gallo & Nair, 2024) are met. With the exclusion of Safety, Security, and Robustness, which are out of scope, the other areas are met as follows.

Appropriateness and Fairness

Recommendations 1,2, and 3, if followed correctly, for an industry, ensure that the data that is selected within curation, utilised by the model following preprocessing and the model's purpose and aim are all appropriate and fair for a given industry by outlining the data that is being used and why.

Transparency and Explainability

Recommendations 1, 2, 3, 4, and 5 ensure that all elements used and produced by the model are explainable, allowing companies to demonstrate why particular data features are utilised and how they impact the decisions and outputs of the models. In doing so, this maintains the transparency of the models and their choices without

relying on traditional, less effective models.

Accountability and Governance

All recommendations enable a clear log of decisions and choices throughout the process, but with the addition of recommendation 6, this ensures that people within a company are accountable for actions that are taken and ensures that these individuals are aware of what is being done and how, as they will be the approvers, who are therefore subject to regulatory or legal ramifications.

Contestability and Redress

The combination of Recommendations 1, 2, 3, 4, and 5 ensures that choices and decisions can be explained to anyone impacted by decisions in a clear manner that they can understand. This will enable customers, regulators, or auditors to contest decisions, as they will be able to provide clear reasons why a decision is incorrect. This, combined with recommendation 6 will allow for models or data to be adjusted and remediation steps to be taken to resolve issues.

Based on the belief that the recommendations outlined in Chapter 5.1 address and provide solutions to adhere to the above framework, it is the belief of the author that Objective 3 has also been achieved.

5.3 Contribution to the Field

As the use of advanced AI methods, particularly Deep Learning, continues to expand and show benefits across various industries, the demand for safe and trustworthy approaches that comply with both current and future regulations will also increase. This investigation aims to make a significant contribution to how regulated industries perceive and utilise Deep Learning techniques. It has aimed to achieve this by:

- Providing examples of how Deep Learning models can improve accuracy performance compared to industry standard methods.
- Enhance trust, transparency, and understanding in Deep Learning through
 - the use of XAI techniques with examples of how to implement them.
 - a set of guidelines/ a framework that companies can utilise that would highlight how best to utilise the XAI techniques to adhere to regulatory requirements.

In completing the points stated above and the objectives laid out in this investigation it is the belief that this will facilitate greater adoption of Deep Learning techniques and therefore improve companies' performances; have a positive economic impact and build trust within the public in how both these techniques are being used, as well as in the decisions that impact their lives, which are being made using the best techniques in a transparent and understandable manner, allowing them to address negative factors.

5.4 Critical Evaluation and Conclusions

This investigation aimed to understand the current limitations that were preventing the utilisation of Deep Learning techniques in various regulated industries, even though there is a general understanding that these techniques offer increased performance.

There were various blockers to the uptake of Deep Learning techniques, mainly surrounding the data usage for the models and the explainability of the models. There is a general lack of trust in these techniques due to a lack of understanding and the perception that they are “black boxes”. As such, this investigation focused on being transparent in how data was prepared, how the model parameters were selected, and lastly, utilising explainable AI techniques to resolve the black box element of Deep Learning Techniques.

Whilst this investigation has not developed a new Deep Learning Technique or method of explaining them, it is the author's belief that this investigation has significantly contributed to bridging the gap between theoretical/ academic work and how that can be utilised in a practical industry setting.

Whilst a practical example has been provided in this investigation, due to the limitations of time and resources on this investigations limiting it to focusing on Financial Services it could not be expanding into other industries and Deep Learning capabilities such as the use of NLP and copywrite infringement when utilising data sources or the ethical considerations that come with using deep learning models with fields like medic, law or automotive industry.

However, throughout this investigation it has demonstrated the value in utilising Deep Learning and that there is the capability to adhere to the regulations and there goal of building consumer trust whilst not stifling progress it is the belief of this author that as industry begins to utilise this technology whilst adhering to the goal of building explainable and trusted methods, that this will benefit both the industries, consumers and increase funding for future research into this field, enabling new and advance deep learning techniques without the mistrust or lack of understanding.

Chapter 6

Future work

This investigation aimed to demonstrate not only the benefits of using deep learning algorithms over widely used traditional models, but also how these models can be applied in adherence to regulations. Whilst this investigation did not focus on the maximisation of deep learning performance, it is worth acknowledging that ongoing advancements in Deep Learning technology could both make models more complex and potentially less naturally explainable. However, based on the trend of explainable AI (google, 2025) There is evidence to suggest that the focus on explainable AI will lead to new advancements, dispelling the notion that Deep Learning Models are black boxes.

However, the focus of this investigation was not on Deep Learning or XAI in isolation, but on how they can be used together within regulated industries. While the performance of the models can be evaluated on a quantitative basis, the explainability factor of the model is subjective to each person, and their understanding of models can influence how well they understand the global explanation. Additionally, having a basic understanding of what the features are has a major impact on the understanding of the explanation. There are methods that can be used to measure someone's understanding that could be used to quantify whether the methods used were truly explained in a way that someone could understand. One such method is explained in (Sands, Parker, Hedgeland, Jordan, & Galloway, 2018).

Another area where the work conducted by this investigation could be further expanded is to validate that the methods and blueprint laid out can span different industries and data types. While this investigation had to be limited to one industry due to time constraints and resources, Financial Services was chosen due to the authors' existing experience in the field. However, future work could expand this to other industries like Law and medicine, where Deep learning techniques with explainable AI could be used to explain sentencing decisions in the judiciary that are currently being assessed (Brunette, 2024). Or to analyse medical scans and explain the reasons it has classified an image in a particular way (National Cancer Institute,

2024). These are just some examples of where getting the best performance is imperative, but being kept accountable and being able to explain decisions is vital, as these are life changing decisions and require heavy regulation, which is why it is hoped that this investigation can help towards utilising Deep Learning to get the best performance while adhering to these regulations.

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Appendix A

Preprocessing data.

Features dropped due to not being available at the application stage

- issue_d
- hardship_flag
- hardship_type
- hardship_reason
- hardship_status
- deferral_term
- hardship_amount
- hardship_start_date
- hardship_end_date
- payment_plan_start_date
- hardship_length
- hardship_dpd
- hardship_loan_status
- orig_projected_additional_accrued_interest
- hardship_payoff_balance_amount
- hardship_last_payment_amount
- disbursement_method
- debt_settlement_flag
- debt_settlement_flag_date
- settlement_status
- settlement_date
- settlement_amount
- settlement_percentage
- settlement_term
- total_rec_int
- total_rec_late_fee
- recoveries
- collection_recovery_fee
- last_pymnt_d
- last_pymnt_amnt
- total_rec_prncp

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- total_pymnt
- total_pymnt_inv
- funded_amnt_inv
- funded_amnt
- delinq_amnt

Features with high% NULL values were dropped due to not providing additional information

- id
- member_id
- url
- next_pymnt_d
- desc
- sec_app_mths_since_last_major_derog
- sec_app_revol_util
- revol_bal_joint
- sec_app_earliest_cr_line
- sec_app_inq_last_6mths
- sec_app_mort_acc
- sec_app_open_acc
- sec_app_open_act_il
- sec_app_num_rev_accts
- sec_app_chargeoff_within_12_mths
- sec_app_collections_12_mths_ex_med
- verification_status_joint
- dti_joint
- annual_inc_joint

Rows with NULL values in the below features that have no clear substitute have been deleted.

- emp_title
- emp_length
- title
- last_pymnt_d
- last_credit_pull_d
- zip_code

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- earliest_cr_line

Rows with NULL values in numerical features have been updated to 0, with reasons given for features with greater than 50% NULL values below.

- never had a public record:
 - mths_since_last_record
- never had a delinquency
 - mths_since_recent_bc_dlq
 - mths_since_last_delinq
 - mths_since_recent_revol_delinq
- never been past due
 - mths_since_last_major_derog
- has no credit balances
 - il_util
 - all_util
 - total_bal_il
 - open_rv_12m
 - open_rv_24m
 - max_bal_bc
- has no previous accounts or inquiries
 - mths_since_rcnt_il
 - open_acc_6m
 - total_cu_tl
 - inq_last_12m
 - open_act_il
 - open_il_12m
 - open_il_24m
 - inq_fi

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- Emp_title was then removed due to being a free text field
- redundant features were removed
 - grade when sub-grade is a more segregated version.
 - title as purpose is a consolidated version.

Categorical data mapping to numeric equivalents

Months

- 36 months = 36
- 60 months = 60

emp_length

- < 1 year = 0
- 1 year = 1
- 2 years = 2
- 3 years = 3
- 4 years = 4
- 5 years = 5
- 6 years = 6
- 7 years = 7
- 8 years = 8
- 9 years = 9
- 10+ years = 10

Appendix B

Column definitions that are shown in the results section

Feature Abbreviations	Feature Details
acc_now_delinq	The number of accounts on which the borrower is now delinquent.
acc_open_past_24mths	Number of trades opened in past 24 months.
addr_state	The state provided by the borrower in the loan application
all_util	Balance to credit limit on all trades
annual_inc	The self-reported annual income provided by the borrower during registration.
annual_inc_joint	The combined self-reported annual income provided by the co-borrowers during registration
application_type	Indicates whether the loan is an individual application or a joint application with two co-borrowers
avg_cur_bal	Average current balance of all accounts
bc_open_to_buy	Total open to buy on revolving bankcards.
bc_util	Ratio of total current balance to high credit/credit limit for all bankcard accounts.
chargeoff_within_12_mths	Number of charge-offs within 12 months
collection_recovery_fee	post charge off collection fee
collections_12_mths_ex_med	Number of collections in 12 months excluding medical collections
delinq_2yrs	The number of 30+ days past-due incidences of delinquency in the borrower's credit file for the past 2 years
delinq_amnt	The past-due amount owed for the accounts on which the borrower is now delinquent.
desc	Loan description provided by the borrower
dti	A ratio calculated using the borrower's total monthly debt payments on the total debt

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	obligations, excluding mortgage and the requested LC loan, divided by the borrower's self-reported monthly income.
dti_joint	A ratio calculated using the co-borrowers' total monthly payments on the total debt obligations, excluding mortgages and the requested LC loan, divided by the co-borrowers' combined self-reported monthly income
earliest_cr_line	The month the borrower's earliest reported credit line was opened
emp_length	Employment length in years. Possible values are between 0 and 10 where 0 means less than one year and 10 means ten or more years.
emp_title	The job title supplied by the Borrower when applying for the loan.*
fico_range_high	The upper boundary range the borrower's FICO at loan origination belongs to.
fico_range_low	The lower boundary range the borrower's FICO at loan origination belongs to.
funded_amnt	The total amount committed to that loan at that point in time.
funded_amnt_inv	The total amount committed by investors for that loan at that point in time.
grade	LC assigned loan grade
home_ownership	The home ownership status provided by the borrower during registration or obtained from the credit report. Our values are: RENT, OWN, MORTGAGE, OTHER
id	A unique LC assigned ID for the loan listing.
il_util	Ratio of total current balance to high credit/credit limit on all install acct
initial_list_status	The initial listing status of the loan. Possible values are – W, F
inq-fi	Number of personal finance inquiries
inq_last_12m	Number of credit inquiries in past 12 months
inq_last_6mths	The number of inquiries in past 6 months (excluding auto and mortgage inquiries)
installment	The monthly payment owed by the borrower if the loan originates.
int_rate	Interest Rate on the loan
issue_d	The month which the loan was funded
last_credit_pull_d	The most recent month LC pulled credit for this loan

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last_fico_range_high	The upper boundary range the borrower's last FICO pulled belongs to.
last_fico_range_low	The lower boundary range the borrower's last FICO pulled belongs to.
last_pymnt_amnt	Last total payment amount received
last_pymnt_d	Last month payment was received
loan_amnt	The listed amount of the loan applied for by the borrower. If at some point in time, the credit department reduces the loan amount, then it will be reflected in this value.
loan_status	Current status of the loan
max_bal_bc	Maximum current balance owed on all revolving accounts
member_id	A unique LC assigned Id for the borrower member.
mo_sin_old_il_acct	Months since oldest bank installment account opened
mo_sin_old_rev_tl_op	Months since oldest revolving account opened
mo_sin_rcnt_rev_tl_op	Months since most recent revolving account opened
mo_sin_rcnt_tl	Months since most recent account opened
mort_acc	Number of mortgage accounts.
mths_since_last_delinq	The number of months since the borrower's last delinquency.
mths_since_last_major_d erog	Months since most recent 90-day or worse rating
mths_since_last_record	The number of months since the last public record.
mths_since_rcnt_il	Months since most recent installment accounts opened
mths_since_recent_bc	Months since most recent bankcard account opened.
mths_since_recent_bc_dl q	Months since most recent bankcard delinquency
mths_since_recent_inq	Months since most recent inquiry.
mths_since_recent_revol_ delinq	Months since most recent revolving delinquency.
next_pymnt_d	Next scheduled payment date
num_accts_ever_120_pd	Number of accounts ever 120 or more days past due
num_actv_bc_tl	Number of currently active bankcard accounts
num_actv_rev_tl	Number of currently active revolving trades
num_bc_sats	Number of satisfactory bankcard accounts

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num_bc_tl	Number of bankcard accounts
num_il_tl	Number of installment accounts
num_op_rev_tl	Number of open revolving accounts
num_rev_accts	Number of revolving accounts
num_rev_tl_bal_gt_0	Number of revolving trades with balance >0
num_sats	Number of satisfactory accounts
num_tl_120dpd_2m	Number of accounts currently 120 days past due (updated in past 2 months)
num_tl_30dpd	Number of accounts currently 30 days past due (updated in past 2 months)
num_tl_90g_dpd_24m	Number of accounts 90 or more days past due in last 24 months
num_tl_op_past_12m	Number of accounts opened in past 12 months
open_acc	The number of open credit lines in the borrower's credit file.
open_acc_6m	Number of open trades in last 6 months
open_il_12m	Number of installment accounts opened in past 12 months
open_il_24m	Number of installment accounts opened in past 24 months
open_act_il	Number of currently active installment trades
open_rv_12m	Number of revolving trades opened in past 12 months
open_rv_24m	Number of revolving trades opened in past 24 months
out_prncp	Remaining outstanding principal for total amount funded
out_prncp_inv	Remaining outstanding principal for portion of total amount funded by investors
pct_tl_nvr_dlq	Percent of trades never delinquent
percent_bc_gt_75	Percentage of all bankcard accounts > 75% of limit.
policy_code	publicly available policy_code=1 new products not publicly available policy_code=2
pub_rec	Number of derogatory public records
pub_rec_bankruptcies	Number of public record bankruptcies
purpose	A category provided by the borrower for the loan request.
pymnt_plan	Indicates if a payment plan has been put in place for the loan
recoveries	post charge off gross recovery

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revol_bal	Total credit revolving balance
revol_util	Revolving line utilization rate, or the amount of credit the borrower is using relative to all available revolving credit.
sub_grade	LC assigned loan subgrade
tax_liens	Number of tax liens
term	The number of payments on the loan. Values are in months and can be either 36 or 60.
title	The loan title provided by the borrower
tot_coll_amt	Total collection amounts ever owed
tot_cur_bal	Total current balance of all accounts
tot_hi_cred_lim	Total high credit/credit limit
total_acc	The total number of credit lines currently in the borrower's credit file
total_bal_ex_mort	Total credit balance excluding mortgage
total_bal_il	Total current balance of all installment accounts
total_bc_limit	Total bankcard high credit/credit limit
total_cu_tl	Number of finance trades
total_il_high_credit_limit	Total installment high credit/credit limit
total_pymnt	Payments received to date for total amount funded
total_pymnt_inv	Payments received to date for portion of total amount funded by investors
total_rec_int	Interest received to date
total_rec_late_fee	Late fees received to date
total_rec_prncp	Principal received to date
total_rev_hi_lim	Total revolving high credit/credit limit
url	URL for the LC page with listing data.
verification_status	Indicates if income was verified by LC, not verified, or if the income source was verified
verified_status_joint	Indicates if the co-borrowers' joint income was verified by LC, not verified, or if the income source was verified
zip_code	The first 3 numbers of the zip code provided by the borrower in the loan application.
revol_bal_joint	Sum of revolving credit balance of the co-borrowers, net of duplicate balances
sec_app_fico_range_low	FICO range (high) for the secondary applicant
sec_app_fico_range_high	FICO range (low) for the secondary applicant

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sec_app_earliest_cr_line	Earliest credit line at time of application for the secondary applicant
sec_app_inq_last_6mths	Credit inquiries in the last 6 months at time of application for the secondary applicant
sec_app_mort_acc	Number of mortgage accounts at time of application for the secondary applicant
sec_app_open_acc	Number of open trades at time of application for the secondary applicant
sec_app_revol_util	Ratio of total current balance to high credit/credit limit for all revolving accounts
sec_app_open_act_il	Number of currently active installment trades at time of application for the secondary applicant
sec_app_num_rev_accts	Number of revolving accounts at time of application for the secondary applicant
sec_app_chargeoff_within_12_mths	Number of charge-offs within last 12 months at time of application for the secondary applicant
sec_app_collections_12_mths_ex_med	Number of collections within last 12 months excluding medical collections at time of application for the secondary applicant
sec_app_mths_since_last_major_derog	Months since most recent 90-day or worse rating at time of application for the secondary applicant
hardship_flag	Flags whether or not the borrower is on a hardship plan
hardship_type	Describes the hardship plan offering
hardship_reason	Describes the reason the hardship plan was offered
hardship_status	Describes if the hardship plan is active, pending, canceled, completed, or broken
deferral_term	Amount of months that the borrower is expected to pay less than the contractual monthly payment amount due to a hardship plan
hardship_amount	The interest payment that the borrower has committed to make each month while they are on a hardship plan
hardship_start_date	The start date of the hardship plan period
hardship_end_date	The end date of the hardship plan period
payment_plan_start_date	The day the first hardship plan payment is due. For example, if a borrower has a hardship plan period of 3 months, the start date is the start of the three-month period in which the borrower is allowed to make interest-only payments.
hardship_length	The number of months the borrower will make smaller payments than normally obligated due to a hardship plan
hardship_dpd	Account days past due as of the hardship plan start

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	date
hardship_loan_status	Loan Status as of the hardship plan start date
orig_projected_additional_accrued_interest	The original projected additional interest amount that will accrue for the given hardship payment plan as of the Hardship Start Date. This field will be null if the borrower has broken their hardship payment plan.
hardship_payoff_balance_amount	The payoff balance amount as of the hardship plan start date
hardship_last_payment_amount	The last payment amount as of the hardship plan start date
disbursement_method	The method by which the borrower receives their loan. Possible values are: CASH, DIRECT_PAY
debt_settlement_flag	Flags whether or not the borrower, who has charged-off, is working with a debt-settlement company.
debt_settlement_flag_date	The most recent date that the Debt Settlement Flag has been set
settlement_status	The status of the borrower's settlement plan. Possible values are: COMPLETE, ACTIVE, BROKEN, CANCELLED, DENIED, DRAFT
settlement_date	The date that the borrower agrees to the settlement plan
settlement_amount	The loan amount that the borrower has agreed to settle for
settlement_percentage	The settlement amount as a percentage of the payoff balance amount on the loan
settlement_term	The number of months that the borrower will be on the settlement plan

Appendix C

Raw results output

```
Epoch 1/10
50461/50461 ————— 105s 2ms/step - AUC: 0.9217 - loss: 0.2443 - val_AUC: 0.9960 - val_loss: 0.0782
Epoch 2/10
50461/50461 ————— 98s 2ms/step - AUC: 0.9634 - loss: 0.1696 - val_AUC: 0.9959 - val_loss: 0.0878
Epoch 3/10
50461/50461 ————— 101s 2ms/step - AUC: 0.9658 - loss: 0.1620 - val_AUC: 0.9962 - val_loss: 0.0935
Epoch 4/10
50461/50461 ————— 102s 2ms/step - AUC: 0.9668 - loss: 0.1593 - val_AUC: 0.9967 - val_loss: 0.0862
Epoch 5/10
50461/50461 ————— 101s 2ms/step - AUC: 0.9675 - loss: 0.1557 - val_AUC: 0.9962 - val_loss: 0.0830
Epoch 6/10
50461/50461 ————— 101s 2ms/step - AUC: 0.9676 - loss: 0.1547 - val_AUC: 0.9963 - val_loss: 0.0894
Epoch 7/10
50461/50461 ————— 101s 2ms/step - AUC: 0.9691 - loss: 0.1530 - val_AUC: 0.9952 - val_loss: 0.0816
Epoch 8/10
50461/50461 ————— 101s 2ms/step - AUC: 0.9697 - loss: 0.1502 - val_AUC: 0.9951 - val_loss: 0.0838
Epoch 9/10
50461/50461 ————— 102s 2ms/step - AUC: 0.9699 - loss: 0.1514 - val_AUC: 0.9957 - val_loss: 0.0892
Epoch 10/10
50461/50461 ————— 102s 2ms/step - AUC: 0.9715 - loss: 0.1469 - val_AUC: 0.9966 - val_loss: 0.0838
```

Figure C.1 Raw output of Learning evolution of DNN

Investigation into how to Increase Deep Learning Utilisation in Regulated Industries

```
Epoch 1/20
7776/7776 ----- 22s 2ms/step - AUC: 0.6054 - loss: 0.5010 - val_AUC: 0.7060
Epoch 2/20
7776/7776 ----- 18s 2ms/step - AUC: 0.6893 - loss: 0.4566 - val_AUC: 0.7065 - val_loss: 0.4511
Epoch 3/20
7776/7776 ----- 17s 2ms/step - AUC: 0.6979 - loss: 0.4509 - val_AUC: 0.7106 - val_loss: 0.4486
Epoch 4/20
7776/7776 ----- 17s 2ms/step - AUC: 0.6986 - loss: 0.4506 - val_AUC: 0.7111 - val_loss: 0.4521
Epoch 5/20
7776/7776 ----- 18s 2ms/step - AUC: 0.7034 - loss: 0.4501 - val_AUC: 0.7127 - val_loss: 0.4463
Epoch 6/20
7776/7776 ----- 18s 2ms/step - AUC: 0.6994 - loss: 0.4492 - val_AUC: 0.7144 - val_loss: 0.4483
Epoch 7/20
7776/7776 ----- 18s 2ms/step - AUC: 0.7039 - loss: 0.4511 - val_AUC: 0.7153 - val_loss: 0.4461
Epoch 8/20
7776/7776 ----- 18s 2ms/step - AUC: 0.7037 - loss: 0.4472 - val_AUC: 0.7143 - val_loss: 0.4508
Epoch 9/20
7776/7776 ----- 18s 2ms/step - AUC: 0.7043 - loss: 0.4480 - val_AUC: 0.7154 - val_loss: 0.4450
Epoch 10/20
7776/7776 ----- 18s 2ms/step - AUC: 0.7064 - loss: 0.4467 - val_AUC: 0.7154 - val_loss: 0.4494
Epoch 11/20
7776/7776 ----- 18s 2ms/step - AUC: 0.7092 - loss: 0.4454 - val_AUC: 0.7155 - val_loss: 0.4465
Epoch 12/20
7776/7776 ----- 18s 2ms/step - AUC: 0.7061 - loss: 0.4462 - val_AUC: 0.7154 - val_loss: 0.4462
Epoch 13/20
7776/7776 ----- 18s 2ms/step - AUC: 0.7071 - loss: 0.4472 - val_AUC: 0.7163 - val_loss: 0.4494
Epoch 14/20
7776/7776 ----- 18s 2ms/step - AUC: 0.7089 - loss: 0.4450 - val_AUC: 0.7164 - val_loss: 0.4465
Epoch 15/20
7776/7776 ----- 18s 2ms/step - AUC: 0.7076 - loss: 0.4463 - val_AUC: 0.7161 - val_loss: 0.4470
Epoch 16/20
7776/7776 ----- 17s 2ms/step - AUC: 0.7083 - loss: 0.4450 - val_AUC: 0.7159 - val_loss: 0.4453
Epoch 17/20
7776/7776 ----- 16s 2ms/step - AUC: 0.7105 - loss: 0.4437 - val_AUC: 0.7161 - val_loss: 0.4463
Epoch 18/20
7776/7776 ----- 17s 2ms/step - AUC: 0.7111 - loss: 0.4458 - val_AUC: 0.7172 - val_loss: 0.4456
Epoch 19/20
7776/7776 ----- 17s 2ms/step - AUC: 0.7064 - loss: 0.4469 - val_AUC: 0.7169 - val_loss: 0.4458
Epoch 20/20
7776/7776 ----- 18s 2ms/step - AUC: 0.7076 - loss: 0.4469 - val_AUC: 0.7170 - val_loss: 0.4465
```

Figure C.2 Raw output of Learning evolution of DNN on subset dataset

Investigation into how to Increase Deep Learning Utilisation in Regulated Industries

```
Train Result:
=====
Accuracy Score: 81.07%

CLASSIFICATION REPORT:

```

	0.0	1.0	accuracy	macro avg	weighted avg
precision	0.535657	0.821350	0.810676	0.678503	0.766500
recall	0.104245	0.978528	0.810676	0.541387	0.810676
f1-score	0.174526	0.893076	0.810676	0.533801	0.755123
support	47772.000000	201055.000000	0.810676	248827.000000	248827.000000

```

Confusion Matrix:
[[ 4980 42792]
 [ 4317 196738]]

Test Result:
=====
Accuracy Score: 80.89%

CLASSIFICATION REPORT:

```

	0.0	1.0	accuracy	macro avg	weighted avg
precision	0.523172	0.819972	0.808946	0.671572	0.762756
recall	0.100821	0.978055	0.808946	0.539438	0.808946
f1-score	0.169062	0.892064	0.808946	0.530563	0.752687
support	23626.000000	98931.000000	0.808946	122557.000000	122557.000000

```

Confusion Matrix:
[[ 2382 21244]
 [ 2171 96760]]
```

Figure C.3 DNN Subset model metrics

```
Train Result:
=====
Accuracy Score: 100.00%

CLASSIFICATION REPORT:

```

	0.0	1.0	accuracy	macro avg	weighted avg
precision	1.000000	0.999985	0.999988	0.999993	0.999988
recall	0.999937	1.000000	0.999988	0.999969	0.999988
f1-score	0.999969	0.999993	0.999988	0.999981	0.999988
support	47772.000000	201055.000000	0.999988	248827.000000	248827.000000

```

Confusion Matrix:
[[ 47769    3]
 [    0 201055]]

Test Result:
=====
Accuracy Score: 80.85%

CLASSIFICATION REPORT:

```

	0.0	1.0	accuracy	macro avg	weighted avg
precision	0.523605	0.816271	0.808481	0.669938	0.759852
recall	0.072293	0.984292	0.808481	0.528293	0.808481
f1-score	0.127046	0.892442	0.808481	0.509744	0.744892
support	23626.000000	98931.000000	0.808481	122557.000000	122557.000000

```

Confusion Matrix:
[[ 1708 21918]
 [ 1554 97377]]
```

Figure C.4 Random Forest Subset model metrics

Investigation into how to Increase Deep Learning Utilisation in Regulated Industries

```
Train Result:
=====
Accuracy Score: 81.01%

CLASSIFICATION REPORT:

```

	0.0	1.0	accuracy	macro avg	weighted avg
precision	0.559041	0.814715	0.810129	0.686878	0.765628
recall	0.052227	0.990212	0.810129	0.521219	0.810129
f1-score	0.095530	0.893931	0.810129	0.494731	0.740647
support	47772.000000	201055.000000	0.810129	248827.000000	248827.000000

```

Confusion Matrix:
[[ 2495 45277]
 [ 1968 199087]]

Test Result:
=====
Accuracy Score: 80.91%

CLASSIFICATION REPORT:

```

	0.0	1.0	accuracy	macro avg	weighted avg
precision	0.550262	0.814025	0.809101	0.682144	0.763178
recall	0.053289	0.989599	0.809101	0.521444	0.809101
f1-score	0.097168	0.893266	0.809101	0.495217	0.739798
support	23626.000000	98931.000000	0.809101	122557.000000	122557.000000

```

Confusion Matrix:
[[ 1259 22367]
 [ 1029 97902]]
```

Figure C.5 Logistic Regression Subset model metrics

Appendix D

Full versions of smaller tables

Industry	Regulator(s)
Financial Services	Financial Conduct Authority (FCA), Prudential Regulation Authority (PRA)
Energy	Ofgem (Office of Gas and Electricity Markets)
Telecommunications	Ofcom (Office of Communications)
Water	Ofwat (Water Services Regulatory Authority)
Rail	ORR (Office of Rail and Road)
Healthcare	Care Quality Commission (CQC)
Law	Legal Services Board (LSB)
Environment	Environment Agency

Table D.1 Examples of regulated industries and their regulator(s) in the UK

model	AUC	F1-score	ACC	Cohen - Kappa	Matthews	precision	recall	mean	rank
DNN-5	0.798	0.688	0.874	0.382	0.387	0.683	0.700	0.6448	1
DNN-4	0.799	0.684	0.872	0.383	0.387	0.682	0.705	0.6446	2
DNN-2	0.800	0.685	0.868	0.378	0.382	0.664	0.723	0.6431	3
DNN-3	0.799	0.689	0.871	0.378	0.381	0.670	0.711	0.6428	4

Investigation into how to Increase Deep Learning Utilisation in Regulated Industries

SVM	0.81 3	0.62 0	0.83 2	0.341	0.366	0.599	0.74 4	0.616 4	5
LLR	0.80 9	0.60 4	0.78 9	0.286	0.316	0.584	0.74 6	0.590 7	6
RF	0.76 0	0.61 2	0.79 5	0.277	0.297	0.594	0.70 0	0.576 5	7
ABD T	0.78 6	0.57 5	0.76 3	0.260	0.294	0.570	0.74 1	0.569 8	8
BNB	0.77 4	0.58 7	0.76 0	0.251	0.282	0.574	0.73 3	0.565 9	9

Table D.2 Ranked Normalised Scores for Each Machine Learning Algorithm by Metric (Average over Eight Datasets for the Test Dataset) (Korotcov, Tkachenko, Russo, & Ekins, 2017)

Abbreviations: (DNN-X) DNN with X hidden layers, (SVM) Support Vector Machines, (LLR) Logistic linear regression, (RF) Random Forest, (ABDT) AdaBoost Decision Trees, (BNB) Bernoulli Naive Bayes. (Korotcov, Tkachenko, Russo, & Ekins, 2017)

Appendix E

Code

The code has been submitted separately with outputs included but a code of the raw code is included below

```
!pip install tensorflow[and-cuda]
!pip install tensorflow
!pip install scikit-learn
!pip install lime
!pip install shap
!pip install seaborn
!pip install hvplot
!pip install researchpy
!pip install utils
!pip install xgboost
```

Run all the imports and environment set up

```
import pandas as pd
import numpy as np
import seaborn as sns
import scipy.stats as stats
from scipy.stats import chi2_contingency
import matplotlib.pyplot as plt
import hvplot.pandas
# import researchpy as rp
from sklearn import preprocessing
from sklearn.model_selection import train_test_split, RandomizedSearchCV
from sklearn.preprocessing import MinMaxScaler, StandardScaler, OneHotEncoder,
LabelEncoder, OrdinalEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from lime.lime_tabular import LimeTabularExplainer
import shap
import utils
shap.initjs()
pd.options.display.max_rows = 4000
```

Model specific imports

```
import tensorflow as tf
from tensorflow.keras.models import Sequential, Model
from tensorflow.keras.layers import Dense, Dropout, BatchNormalization
from tensorflow.keras.optimizers import Adam, AdamW
from tensorflow.keras.metrics import AUC

from sklearn.model_selection import train_test_split, RandomizedSearchCV
from sklearn.preprocessing import MinMaxScaler, Normalizer

from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression

from xgboost import XGBClassifier

from sklearn.metrics import (accuracy_score, confusion_matrix, classification_report,
                             roc_auc_score,
                             roc_curve, auc, ConfusionMatrixDisplay, RocCurveDisplay, matthews_corrcoef)
```

Loading Dataset

```
data = pd.read_csv("Datasets/loan.csv", skipinitialspace=True)
print(data.shape)
data.head()

print(data.info())
data['loan_status'].value_counts()
data = data[data['loan_status'].isin(['Fully Paid', 'Charged Off'])]
data['loan_status'].value_counts()

data['loan_status'].value_counts().hvplot.bar(
    title="Loan_Status Count", xlabel='Loan_Status', ylabel='Count', width=1000, height=500,
    yformatter='%.0f', fontsize=2)
```

Converting target variable to boolean

Old value	New value
-----------	-----------

Fully Paid	1
------------	---

Investigation into how to Increase Deep Learning Utilisation in Regulated Industries

Old value New value

Charged Off 0

```
data['loan_status'] = data.loan_status.map({'Fully Paid':1, 'Charged Off':0})  
print(data.loan_status.unique())
```

```
float_cols = data.select_dtypes("float64")  
cat_cols = data.select_dtypes("object")
```

```
float_headers = list(float_cols.columns.values)  
cat_headers = list(cat_cols.columns.values)
```

```
float_headers.append('loan_status')
```

Dropping features that would not be present at application:

- issue_d
- hardship_flag
- hardship_type
- hardship_reason
- hardship_status
- deferral_term
- hardship_amount
- hardship_start_date
- hardship_end_date
- payment_plan_start_date
- hardship_length
- hardship_dpd
- hardship_loan_status
- orig_projected_additional_accrued_interest
- hardship_payoff_balance_amount
- hardship_last_payment_amount
- disbursement_method
- debt_settlement_flag
- debt_settlement_flag_date
- settlement_status
- settlement_date
- settlement_amount
- settlement_percentage
- settlement_term

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- total_rec_late_fee
- recoveries
- collection_recovery_fee
- last_pymnt_d
- last_pymnt_amnt
- total_rec_prncp
- total_pymnt
- total_pymnt_inv
- funded_amnt_inv
- funded_amnt
- delinq_amnt
- out_prncp_inv
- out_prncp

```
data.drop('issue_d', axis=1, inplace=True)
data.drop('hardship_flag', axis=1, inplace=True)
data.drop('hardship_type', axis=1, inplace=True)
data.drop('hardship_reason', axis=1, inplace=True)
data.drop('hardship_status', axis=1, inplace=True)
data.drop('deferral_term', axis=1, inplace=True)
data.drop('hardship_amount', axis=1, inplace=True)
data.drop('hardship_start_date', axis=1, inplace=True)
data.drop('hardship_end_date', axis=1, inplace=True)
data.drop('payment_plan_start_date', axis=1, inplace=True)
data.drop('hardship_length', axis=1, inplace=True)
data.drop('hardship_dpd', axis=1, inplace=True)
data.drop('hardship_loan_status', axis=1, inplace=True)
data.drop('orig_projected_additional_accrued_interest', axis=1, inplace=True)
data.drop('hardship_payoff_balance_amount', axis=1, inplace=True)
data.drop('hardship_last_payment_amount', axis=1, inplace=True)
data.drop('disbursement_method', axis=1, inplace=True)
data.drop('debt_settlement_flag', axis=1, inplace=True)
data.drop('debt_settlement_flag_date', axis=1, inplace=True)
data.drop('settlement_status', axis=1, inplace=True)
data.drop('settlement_date', axis=1, inplace=True)
data.drop('settlement_amount', axis=1, inplace=True)
data.drop('settlement_percentage', axis=1, inplace=True)
data.drop('settlement_term', axis=1, inplace=True)
data.drop('total_rec_int', axis=1, inplace=True)
data.drop('total_rec_late_fee', axis=1, inplace=True)
data.drop('recoveries', axis=1, inplace=True)
data.drop('collection_recovery_fee', axis=1, inplace=True)
data.drop('last_pymnt_d', axis=1, inplace=True)
data.drop('last_pymnt_amnt', axis=1, inplace=True)
data.drop('total_rec_prncp', axis=1, inplace=True)
data.drop('total_pymnt', axis=1, inplace=True)
```

```
data.drop('total_pymnt_inv', axis=1, inplace=True)
data.drop('funded_amnt_inv', axis=1, inplace=True)
data.drop('funded_amnt', axis=1, inplace=True)
data.drop('delinq_amnt', axis=1, inplace=True)
data.drop('out_prncp_inv', axis=1, inplace=True)
data.drop('out_prncp', axis=1, inplace=True)
```

```
data.head()
```

6 Investigate rows with missing data

```
print(f" The number of rows in the dataset: {len(data)}\n")
# find number of missing values in columns
missing_data_variables = {}
for column in data.columns:
    if data[column].isna().sum() != 0:
        num_missing_rows = data[column].isna().sum()
        percentage = (num_missing_rows / len(data)) * 100
        missing_data_variables.update({column : percentage})

# print(f"{column} : The number of rows with missing values:{num_missing_rows} / The
percentage of rows:{percentage:.2f}%")
missing_data_variables = dict(sorted(missing_data_variables.items(), key=lambda item:
item[1], reverse = True))
for key in missing_data_variables:
    print(key, 'The percentage of missing rows is:', missing_data_variables[key], '%')
# data.dropna(inplace = True)
# print(f"\nThe number of rows in the dataset after rows with missing values are dropped:
{len(data)}")
```

I will drop all columns with 100% missing values as these provide no data

- id
- member_id
- url
- next_pymnt_d

```
data.drop('id', axis=1, inplace=True)
data.drop('member_id', axis=1, inplace=True)
data.drop('url', axis=1, inplace=True)
data.drop('next_pymnt_d', axis=1, inplace=True)
```

I will drop all columns related to joint application or second applicant as these have over 98% missing values so make up a very small population

- sec_app_mths_since_last_major_derog
- sec_app_revol_util
- revol_bal_joint
- sec_app_earliest_cr_line
- sec_app_inq_last_6mths
- sec_app_mort_acc
- sec_app_open_acc
- sec_app_open_act_il
- sec_app_num_rev_accts
- sec_app_chargeoff_within_12_mths
- sec_app_collections_12_mths_ex_med
- verification_status_joint
- dti_joint
- annual_inc_joint

```
data.drop('sec_app_mths_since_last_major_derog', axis=1, inplace=True)
data.drop('sec_app_revol_util', axis=1, inplace=True)
data.drop('revol_bal_joint', axis=1, inplace=True)
data.drop('sec_app_earliest_cr_line', axis=1, inplace=True)
data.drop('sec_app_inq_last_6mths', axis=1, inplace=True)
data.drop('sec_app_mort_acc', axis=1, inplace=True)
data.drop('sec_app_open_acc', axis=1, inplace=True)
data.drop('sec_app_open_act_il', axis=1, inplace=True)
data.drop('sec_app_num_rev_accts', axis=1, inplace=True)
data.drop('sec_app_chargeoff_within_12_mths', axis=1, inplace=True)
data.drop('sec_app_collections_12_mths_ex_med', axis=1, inplace=True)
data.drop('verification_status_joint', axis=1, inplace=True)
data.drop('dti_joint', axis=1, inplace=True)
data.drop('annual_inc_joint', axis=1, inplace=True)
```

```
missing_data_variables = {}
for column in data.columns:
    if data[column].isna().sum() != 0:
        num_missing_rows = data[column].isna().sum()
        percentage = (num_missing_rows / len(data)) * 100
        missing_data_variables.update({column : percentage})
```

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```
# print(f'{column}: The number of rows with missing values:{num_missing_rows} / The  
percentage of rows:{percentage:.2f}%')  
missing_data_variables = dict(sorted(missing_data_variables.items(), key=lambda item:  
item[1], reverse = True))
```

```
float_cols = data.select_dtypes("float64")  
cat_cols = data.select_dtypes("object")
```

```
float_headers = list(float_cols.columns.values)  
cat_headers = list(cat_cols.columns.values)
```

```
print("All categoric variables\n")  
for key in missing_data_variables:  
    if key in cat_headers:  
        print(key, 'The percentage of missing rows is:', missing_data_variables[key], '%')  
  
print("\n\nAll numeric variables\n")  
for key in missing_data_variables:  
    if key in float_headers:  
        print(key, 'The percentage of missing rows is:', missing_data_variables[key], '%')
```

Dropping categoric features with high NULL values/ free text fields

desc Loan description provided by the borrower this is a free text field and has ~91% unique

```
data.drop('desc', axis=1, inplace=True)
```

For the categoric variables with missing values these rows will be deleted as there is no way to accurately know what they would be and a default value could bias the data.

```
print(f" The number of rows in the dataset: {len(data)}\n")  
# data = data.dropna(subset=['emp_title', 'emp_length', 'title', 'last_pymnt_d', 'last_credit_pull_d',  
'zip_code', 'earliest_cr_line'])  
data = data.dropna(subset=['emp_title', 'emp_length', 'title', 'last_credit_pull_d', 'zip_code',  
'earliest_cr_line'])  
print(f" The number of rows in the dataset: {len(data)}\n")
```

```
missing_data_variables = {}  
for column in data.columns:  
    if data[column].isna().sum() != 0:  
        num_missing_rows = data[column].isna().sum()
```

Investigation into how to Increase Deep Learning Utilisation in Regulated Industries

```
percentage = (num_missing_rows / len(data)) * 100
missing_data_variables.update({column : percentage})

# print(f"{column}: The number of rows with missing values:{num_missing_rows} / The
percentage of rows:{percentage:.2f}%")
missing_data_variables = dict(sorted(missing_data_variables.items(), key=lambda item:
item[1], reverse = True))

float_cols = data.select_dtypes("float64")
cat_cols = data.select_dtypes("object")

float_headers = list(float_cols.columns.values)
cat_headers = list(cat_cols.columns.values)

print("All categoric variables\n")
for key in missing_data_variables:
    if key in cat_headers:
        print(key, 'The percentage of missing rows is:', missing_data_variables[key], '%')

print("\n\nAll numeric variables\n")
for key in missing_data_variables:
    if key in float_headers:
        print(key, 'The percentage of missing rows is:', missing_data_variables[key], '%')
```

For the numeric variables with missing values these rows values will be set to 0 as for for these variables it indicates that the applicant may have:

- never had a public record:
 - mths_since_last_record
- never had a delinquency
 - mths_since_recent_bc_dlq
 - mths_since_last_delinq
 - mths_since_recent_revol_delinq
- never been past due
 - mths_since_last_major_derog
- has no credit balances
 - il_util
 - all_util
 - total_bal_il
 - open_rv_12m

- open_rv_24m
- max_bal_bc
- has no previous accounts or inquiries
 - mths_since_rcnt_il
 - open_acc_6m
 - total_cu_tl
 - inq_last_12m
 - open_act_il
 - open_il_12m
 - open_il_24m
 - inq_fi

Examples have been given where more than 50% of the variable is null

```
data = data.fillna(0)
missing_data_variables = {}
for column in data.columns:
    if data[column].isna().sum() != 0:
        num_missing_rows = data[column].isna().sum()
        percentage = (num_missing_rows / len(data)) * 100
        missing_data_variables.update({column : percentage})

# print(f"{column} : The number of rows with missing values:{num_missing_rows} / The
percentage of rows:{percentage:.2f}%")
missing_data_variables = dict(sorted(missing_data_variables.items(), key=lambda item:
item[1], reverse = True))

float_cols = data.select_dtypes("float64")
cat_cols = data.select_dtypes("object")

float_headers = list(float_cols.columns.values)
cat_headers = list(cat_cols.columns.values)

print("All categoric variables\n")
for key in missing_data_variables:
    if key in cat_headers:
        print(key, 'The percentage of missing rows is:', missing_data_variables[key], '%')

print("\n\nAll numeric variables\n")
for key in missing_data_variables:
    if key in float_headers:
        print(key, 'The percentage of missing rows is:', missing_data_variables[key], '%')
```

looking for columns with high number of unique values that could cause overfitting or provide little value

```
col=data.columns

for i in col:
    unique_percentage = (data[i].nunique() / data[i].count())*100
    if unique_percentage > 15:
        print(data.dtypes[i])
        print("Number of unique values in "+ i +" = ", unique_percentage)
```

Dropping features with high unique values/ free text fields

- emp_title ~30% uniques and free text field

```
data.drop('emp_title', axis=1, inplace=True)
```

Dropping redundant columns that are subsets of another columns

I'm dropping grade as it is just the first letter of sub-grade so I can derive all the same information from subgrade but there is still not a high level of uniqueness

```
print(data['grade'].unique())
print(data['sub_grade'].unique())
```

```
data.drop('grade', axis=1, inplace=True)
```

I'm dropping title for the same reason as it is a plain text version of purpose where purpose has already been processed such as combining debt consolidation and consolidation

```
print(data['purpose'].unique())
print(data['title'].unique())
```

```
data.drop('title', axis=1, inplace=True)
```

Drop both zip_code and addr_state as your address is not used for

credit worthiness

but rather a fraud check and as that is not part of any feature or a potential outcome we will not be checking for this

```
print(data['addr_state'].unique())
print(data['zip_code'].unique())
data.drop('addr_state', axis=1, inplace=True)
data.drop('zip_code', axis=1, inplace=True)
```

convert date fields to numeric

```
print([column for column in data.columns if data[column].dtype == object])
data['earliest_cr_line'] = pd.to_datetime(data['earliest_cr_line'])
data['last_credit_pull_d'] = pd.to_datetime(data['last_credit_pull_d'])
data['earliest_cr_line'] = pd.to_numeric(data['earliest_cr_line'])
data['last_credit_pull_d'] = pd.to_numeric(data['last_credit_pull_d'])
```

Mapping variable to clear numeric representative

- I am mapping term to remove 'months' leaving it as 36 and 60 respectively

Old value	New value
-----------	-----------

36 months	36
-----------	----

60 months	60
-----------	----

- I will be converting emp_length into a numeric mapping with years being removed and <1 being 0 and 10+ being 10

Old value	New value
-----------	-----------

< 1 year	0
----------	---

1 year	1
--------	---

2 years	2
---------	---

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Old value New value

3 years 3

4 years 4

5 years 5

6 year 6

7 years 7

8 years 8

9 years 9

10+ years 10

```
dic_terms = {'36 months': 36.0, '60 months': 60.0}
dic_emp_length = {'< 1 year': 0.0, '1 year': 1.0, '2 years': 2.0, '3 years': 3.0, '4 years': 4.0, '5 years': 5.0,
                  '6 years': 6.0, '7 years': 7.0, '8 years': 8.0, '9 years': 9.0, '10+ years': 10.0}
```

```
data['term'] = data.term.map(dic_terms)
data['emp_length'] = data.emp_length.map(dic_emp_length)
print(data.term.unique())
print(data.emp_length.unique())
```

```
data_no_dummies = data
```

Separate target and features

```
features = data.drop('loan_status', axis=1)
target = data['loan_status']
```

Encode categorical features

```
categorical_features = features.select_dtypes(include=['category', 'object']).columns
label_encoders = {}
for col in categorical_features:
    le = OrdinalEncoder()
    features[col] = le.fit_transform(features[col].values.reshape(-1, 1))
    label_encoders[col] = le
```

#create train and test sets

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```
x_train, x_test, y_train, y_test = train_test_split(features, target, test_size=0.33, random_state=0)
x_train_df = x_train
y_train_df = y_train
x_test_df = x_test
y_test_df = y_test
print(x_train)
```

```
scaler = MinMaxScaler()
x_train = scaler.fit_transform(x_train)
x_test = scaler.transform(x_test)
```

```
def print_results(true, pred, train=True):
    if train:
        clf_report = pd.DataFrame(classification_report(true, pred, output_dict=True))
        print("Train Result:\n=====")
        print(f"Accuracy Score: {accuracy_score(true, pred) * 100:.2f}%")
        print("_____")
        print(f"CLASSIFICATION REPORT:\n{clf_report}")
        print("_____")
        print(f"Confusion Matrix: \n {confusion_matrix(true, pred)}\n")
```

```
    elif train==False:
        clf_report = pd.DataFrame(classification_report(true, pred, output_dict=True))
        print("Test Result:\n=====")
        print(f"Accuracy Score: {accuracy_score(true, pred) * 100:.2f}%")
        print("_____")
        print(f"CLASSIFICATION REPORT:\n{clf_report}")
        print("_____")
        print(f"Confusion Matrix: \n {confusion_matrix(true, pred)}\n")
```

```
x_train = np.array(x_train).astype(np.float64)
x_test = np.array(x_test).astype(np.float64)
y_train = np.array(y_train).astype(np.float64)
y_test = np.array(y_test).astype(np.float64)
```

```
def plot_learning_evolution(r):
    plt.figure(figsize=(12, 8))
```

```
    plt.subplot(2, 2, 1)
    plt.plot(r.history['loss'], label='Loss')
    plt.plot(r.history['val_loss'], label='val_Loss')
    plt.title('Loss evolution during trainig')
    plt.legend()
```

```
    plt.subplot(2, 2, 2)
    plt.plot(r.history['AUC'], label='AUC')
    plt.plot(r.history['val_AUC'], label='val_AUC')
    plt.title('AUC score evolution during trainig')
    plt.legend();
```

```
def ann_model(num_columns, num_labels, hidden_layers, dropout_rates, learning_rate):
```

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```
inp = tf.keras.layers.Input(shape=(num_columns,))
x = BatchNormalization()(inp)
x = Dropout(dropout_rates[0])(x)
for i in range(len(hidden_layers)):
    x = Dense(hidden_layers[i], activation='relu')(x)
    x = BatchNormalization()(x)
    x = Dropout(dropout_rates[i + 1])(x)
x = Dense(num_labels, activation='sigmoid')(x)

model = Model(inputs=inp, outputs=x)
model.compile(optimizer=Adam(learning_rate), loss='binary_crossentropy',
metrics=[AUC(name='AUC')])
return model
print(x_train.shape)
print(x_test.shape)
import random
num_columns = x_train.shape[1]
num_labels = 1
num_of_layers = 10
nodes_per_layer = 83

hidden_layers = [nodes_per_layer] * num_of_layers

print(hidden_layers)

dropout_rates = []
for i in range(num_of_layers+1):
    dropout_rates.append(round(random.uniform(0.2, 0.5), 1))
    print(dropout_rates)
learning_rate = 0.005

ANN_model = ann_model(num_columns, num_labels, hidden_layers, dropout_rates,
learning_rate)

r = ANN_model.fit(x_train, y_train, validation_data=(x_test, y_test), epochs= 10, batch_size=16)

plot_learning_evolution(r)

y_train_pred = ANN_model.predict(x_train)
y_test_pred = ANN_model.predict(x_test)
print_results(y_train, y_train_pred.round(), train=True)
print_results(y_test, y_test_pred.round(), train=False)

scores_dict = {
```

```
'ANNs': {
    'Train': roc_auc_score(y_train, ANN_model.predict(x_train), multi_class='ovr'),
    'Test': roc_auc_score(y_test, ANN_model.predict(x_test), multi_class='ovr'),
},
}

print(scores_dict)
```

7 Explaining the ANN

Using lime package to explain the the ANN model

this is completed by building an explainer model that that the same inputs as the ANN model but the target variable is the

```
x_train_df = x_train
y_train_df = y_train
x_test_df = x_test
y_test_df = y_test
# Create the explainer
# data.info()
feature_names = data.columns.tolist()
# print(feature_names)
# # print(feature_names)

# x_test_nd_array = x_test.to_numpy()
print(len(x_test[1]))

explainer = LimeTabularExplainer(x_test,
                                feature_names=feature_names,
                                class_names=['Charged Off', 'Fully Paid'],
                                mode='classification',
                                random_state=42)
```

```
i = 3
```

```
def predict_fn(x):
    # Get the raw predictions from your model
    preds = ANN_model.predict(x)

    if len(preds.shape) == 1 or preds.shape[1] == 1:
        # Return probabilities for both classes
```

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```
    return np.column_stack([1-preds, preds])
else:
    return preds

exp = explainer.explain_instance(x_test[i], predict_fn, num_features=15)

print(y_test[i])

exp.show_in_notebook(show_table=True)

exp.save_to_file('explanation_1.html')


i = 30 # example for charged off
# i = 37 # example for fully paid


def predict_fn(x):
    # Get the raw predictions from your model
    preds = ANN_model.predict(x)

    if len(preds.shape) == 1 or preds.shape[1] == 1:
        # Return probabilities for both classes
        return np.column_stack([1-preds, preds])
    else:
        return preds

exp = explainer.explain_instance(x_test[i], predict_fn, num_features=15)

print(y_test[i])

exp.show_in_notebook(show_table=True)

exp.save_to_file('explanation_1.html')
```

Get the explanation of a single row

```
# Check calculations
exp_weight = [x_test[1] for x_test in exp.as_map()[1]]

print(np.mean(y_test) + sum(exp_weight))
print(y_test[0])


def return_weights(exp):
    """Get weights from LIME explanation object"""
```

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```
exp_list = exp.as_map()[1]
exp_list = sorted(exp_list, key=lambda x_test: x_test[0])
exp_weight = [x_test[1] for x_test in exp_list]

return exp_weight

feature_names.remove('loan_status')
print(len(feature_names))
weights = []

exp = explainer.explain_instance(x_test[0], predict_fn, num_features=15)

# Iterate over first 100 rows
for i in range(0,100):

    # Get explanation
    exp = explainer.explain_instance(x_test[0], predict_fn, num_features=90)

    # Get weights
    exp_weight = return_weights(exp)
    weights.append(exp_weight)

exp.show_in_notebook(show_table=True)

# Create DataFrame
lime_weights = pd.DataFrame(data=weights, columns = feature_names)
```

Get absolute mean of LIME weights

```
abs_mean = lime_weights.abs().mean(axis=0)
abs_mean = pd.DataFrame(data={'feature':abs_mean.index, 'abs_mean':abs_mean})
abs_mean = abs_mean.sort_values('abs_mean')
# print(abs_mean)

abs_mean = abs_mean.tail(15)

# Plot abs mean
fig, ax = plt.subplots(nrows=1, ncols=1,figsize=(8,4))

y_ticks = range(len(abs_mean))
y_labels = abs_mean.feature

plt.barh(y=y_ticks,width=abs_mean.abs_mean,)
plt.yticks(ticks=y_ticks,labels=y_labels,size= 12)
plt.title("Top 15 factors for impacting loan status")
plt.ylabel('Factors')
```

```
plt.xlabel('Absolute Mean Weights',size=15)
```

Using SHAP Package

```
# lime_weights = pd.DataFrame(data=weights, columns = feature_names)
```

```
masker = shap.maskers.Independent(x_train[0:100])
```

```
explainer = shap.Explainer(ANN_model, masker)  
shap_values = explainer(x_train[0:100])
```

```
# Replace SHAP values with LIME weights  
shap_placeholder = explainer(x_train[0:100])  
shap_placeholder.values = np.array(lime_weights)
```

```
explanation = shap.Explanation(  
    values=shap_placeholder.values,  
    base_values=shap_placeholder.base_values,  
    data=shap_placeholder.data,  
    feature_names=feature_names  
)
```

```
shap_explanation = shap.Explanation(  
    values=shap_values,  
    base_values=shap_placeholder.base_values,  
    data=shap_placeholder.data,  
    feature_names=feature_names  
)
```

```
# plt.xlabel("SHAP Values")  
plt.title('SHAP Values of top 9 features and the remaining features for single instance')  
shap.plots.waterfall(shap_values[0],show=False)  
plt.show()
```

```
plt.xlabel("SHAP Values")  
plt.title('SHAP Values of top 9 features and the remaining features for single instance')  
shap.plots.waterfall(shap_values[3],show=False)  
plt.show()
```

```
plt.xlabel("Mean LIME Weight")  
plt.title('Mean LIME Weight of top 9 features and the remaining features for single instance')  
shap.plots.waterfall(explanation[0],show=False)  
plt.show()
```

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```
plt.xlabel("Mean LIME Weight")
plt.title('Mean LIME Weight of top 9 features and the remaining features for single instance')
shap.plots.waterfall(explanation[3],show=False)
plt.show()
```

```
shap.plots.beeswarm(shap_explanation)
```

```
LR_model = LogisticRegression()
LR_model.fit(x_train, y_train) #,validation_data=(x_test, y_test), epochs=20, batch_size=32)
```

```
LR_model.predict(x_test)
LR_model.score(x_test,y_test)
```

```
y_test_pred = LR_model.predict(x_test)
print_results(y_test, y_test_pred.round(), train=False)
```

```
scores_dict['Logistic Regression'] = {
    'Train': roc_auc_score(y_train, LR_model.predict(x_train)),
    'Test': roc_auc_score(y_test, LR_model.predict(x_test)),
}
randomForest_clf = RandomForestClassifier(n_estimators=100)

randomForest_clf.fit(x_train, y_train)
```

```
y_train_pred = randomForest_clf.predict(x_train)
y_test_pred = randomForest_clf.predict(x_test)
print_results(y_train, y_train_pred, train=True)
print_results(y_test, y_test_pred, train=False)
```

```
scores_dict['Random Forest'] = {
    'Train': roc_auc_score(y_train, randomForest_clf.predict(x_train)),
    'Test': roc_auc_score(y_test, randomForest_clf.predict(x_test)),
}
ml_models = {
    'Random Forest': randomForest_clf,
    'Logistic Regression': LR_model,
    'ANNs': ANN_model
}
```

```
for model in ml_models:
    print(f"{model.upper():{50}} roc_auc_score: {roc_auc_score(y_test,
ml_models[model].predict(x_test)):.3f}")
```

```
scores_df = pd.DataFrame(scores_dict)
scores_df.hvplot.barh(
    width=500, height=500,
    xlim=(0.99, 1),
    title="ROC Scores of ML Models", xlabel="ROC Scores", legend='top',
)
```

8 This is a subset of the dataset to compare performance between models.

```
data_2 = pd.read_csv("Datasets/loan_subset.csv", skipinitialspace=True)
print(data_2.shape)
data_2.head()
```

```
data_2['loan_status'].value_counts().hvplot.bar(
    title="Loan Status Totals", xlabel='Loan Status', ylabel='Count',
    width=1000, height=500, yformatter='%0f')
data_2['loan_status'] = data_2.loan_status.map({'Fully Paid':1, 'Charged Off':0})
missing_data_variables = {}
for column in data_2.columns:
    if data_2[column].isna().sum() != 0:
        num_missing_rows = data_2[column].isna().sum()
        percentage = (num_missing_rows / len(data_2)) * 100
        missing_data_variables.update({column : percentage})

# print(f"{column} : The number of rows with missing values:{num_missing_rows} / The
percentage of rows:{percentage:.2f}%")
missing_data_variables = dict(sorted(missing_data_variables.items(), key=lambda item:
item[1], reverse = True))
```

```
float_cols = data_2.select_dtypes("float64")
cat_cols = data_2.select_dtypes("object")
```

```
float_headers = list(float_cols.columns.values)
cat_headers = list(cat_cols.columns.values)
```

```
print("All categoric variables\n")
for key in missing_data_variables:
```

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```
if key in cat_headers:
    print(key, 'The percentage of missing rows is:', missing_data_variables[key], '%')

print("\n\nAll numeric variables\n")
for key in missing_data_variables:
    if key in float_headers:
        print(key, 'The percentage of missing rows is:', missing_data_variables[key], '%')

print(f" The number of rows in the dataset: {len(data_2)}\n")
data_2 = data_2.dropna(subset=['emp_title', 'emp_length', 'title'])
print(f" The number of rows in the dataset: {len(data_2)}\n")

data_2 = data_2.fillna(0)
missing_data_variables = {}
for column in data_2.columns:
    if data_2[column].isna().sum() != 0:
        num_missing_rows = data_2[column].isna().sum()
        percentage = (num_missing_rows / len(data_2)) * 100
        missing_data_variables.update({column : percentage})

# print(f"{column} : The number of rows with missing values:{num_missing_rows} / The
percentage of rows:{percentage:.2f}%")
missing_data_variables = dict(sorted(missing_data_variables.items(), key=lambda item:
item[1], reverse = True))

float_cols = data_2.select_dtypes("float64")
cat_cols = data_2.select_dtypes("object")

float_headers = list(float_cols.columns.values)
cat_headers = list(cat_cols.columns.values)

print("All categoric variables\n")
for key in missing_data_variables:
    if key in cat_headers:
        print(key, 'The percentage of missing rows is:', missing_data_variables[key], '%')

print("\n\nAll numeric variables\n")
for key in missing_data_variables:
    if key in float_headers:
        print(key, 'The percentage of missing rows is:', missing_data_variables[key], '%')
```

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```
data_2.drop('emp_title', axis=1, inplace=True)
data_2.drop('grade', axis=1, inplace=True)
data_2.drop('title', axis=1, inplace=True)
data_2.drop('address', axis=1, inplace=True)
data_2.drop('issue_d', axis=1, inplace=True)
print([column for column in data_2.columns if data_2[column].dtype == object])

dic_terms = {'36 months': 36.0, '60 months': 60.0}
dic_emp_length = {'< 1 year': 0.0, '1 year': 1.0, '2 years': 2.0, '3 years': 3.0, '4 years': 4.0, '5 years': 5.0,
                  '6 years': 6.0, '7 years': 7.0, '8 years': 8.0, '9 years': 9.0, '10+ years': 10.0}

data_2['term'] = data_2.term.map(dic_terms)
data_2['emp_length'] = data_2.emp_length.map(dic_emp_length)
data_2['earliest_cr_line'] = pd.to_datetime(data_2['earliest_cr_line'])
data_2['earliest_cr_line'] = pd.to_numeric(data_2['earliest_cr_line'])
print([column for column in data_2.columns if data_2[column].dtype == object])
features = data_2.drop('loan_status', axis=1)
target = data_2['loan_status']

dummies = ['sub_grade', 'home_ownership', 'verification_status', 'purpose', 'initial_list_status',
            'application_type']
data_2 = pd.get_dummies(data_2, columns=dummies, drop_first=True)
train, test = train_test_split(data_2, test_size=0.33, random_state=0)
x_train_2, y_train_2 = train.drop('loan_status', axis=1), train.loan_status
x_test_2, y_test_2 = test.drop('loan_status', axis=1), test.loan_status
scaler = MinMaxScaler()
x_train_2 = scaler.fit_transform(x_train_2)
x_test_2 = scaler.transform(x_test_2)
x_train_2 = np.array(x_train_2).astype(np.float32)
x_test_2 = np.array(x_test_2).astype(np.float32)
y_train_2 = np.array(y_train_2).astype(np.float32)
y_test_2 = np.array(y_test_2).astype(np.float32)
import random
num_columns = x_train_2.shape[1]
num_labels = 1
num_of_layers = 10
nodes_per_layer = 50

hidden_layers = [nodes_per_layer] * num_of_layers

print(hidden_layers)

dropout_rates = []
for i in range(num_of_layers+1):
    dropout_rates.append(round(random.uniform(0.0, 0.1), 1))
```

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```
print(dropout_rates)
learning_rate = 0.001
```

```
ANN_model = ann_model(num_columns, num_labels, hidden_layers, dropout_rates,
learning_rate)
```

```
r = ANN_model.fit(x_train_2, y_train_2, validation_data=(x_test_2, y_test_2), epochs=10,
batch_size=16)
```

```
plot_learning_evolution(r)
y_train_2_pred = ANN_model.predict(x_train_2)
y_test_2_pred = ANN_model.predict(x_test_2)
print_results(y_train_2, y_train_2_pred.round(), train=True)
print_results(y_test_2, y_test_2_pred.round(), train=False)
```

```
scores_dict = {
    'ANNs': {
        'Train': roc_auc_score(y_train_2, ANN_model.predict(x_train_2), multi_class='ovr'),
        'Test': roc_auc_score(y_test_2, ANN_model.predict(x_test_2), multi_class='ovr'),
    },
}

print(scores_dict)
```

```
rf_clf = RandomForestClassifier(n_estimators=100)
rf_clf.fit(x_train_2, y_train_2)
```

```
y_train_2_pred = rf_clf.predict(x_train_2)
y_test_2_pred = rf_clf.predict(x_test_2)
```

```
print_results(y_train_2, y_train_2_pred, train=True)
print_results(y_test_2, y_test_2_pred, train=False)
```

```
disp = ConfusionMatrixDisplay.from_estimator(
    rf_clf, x_test_2, y_test_2,
    cmap='Blues', values_format='d',
    display_labels=['Charged Off', 'Fully-Paid']
)
```

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```
disp = RocCurveDisplay.from_estimator(rf_clf, x_test_2, y_test_2)
LR_model = LogisticRegression()
LR_model.fit(x_train_2, y_train_2)

y_train_2_pred = LR_model.predict(x_train_2)
y_test_2_pred = LR_model.predict(x_test_2)

print_results(y_train_2, y_train_2_pred, train=True)
print_results(y_test_2, y_test_2_pred, train=False)

disp = ConfusionMatrixDisplay.from_estimator(rf_clf, x_test_2, y_test_2,
                                              cmap='Blues', values_format='d',
                                              display_labels=['Charged Off', 'Fully-Paid'])

disp = RocCurveDisplay.from_estimator(LR_model, x_test_2, y_test_2)
RocCurveDisplay.from_estimator(LR_model, x_test_2, y_test_2, ax=disp.ax_)
scores_dict['Logistic Regression'] = {
    'Train': roc_auc_score(y_train_2, LR_model.predict(x_train_2)),
    'Test': roc_auc_score(y_test_2, LR_model.predict(x_test_2)),
}
scores_dict['Random Forest'] = {
    'Train': roc_auc_score(y_train_2, rf_clf.predict(x_train_2)),
    'Test': roc_auc_score(y_test_2, rf_clf.predict(x_test_2)),
}
scores_df = pd.DataFrame(scores_dict)
scores_df.hvplot.barh(
    width=500, height=400,
    title="ROC Scores of ML Models", xlabel="ROC Scores",
    alpha=1, legend='top'
)

# Create the explainer
feature_names = data_2.columns.tolist()

print(len(x_test[1]))

explainer = LimeTabularExplainer(x_test_2,
                                 feature_names=feature_names,
                                 class_names=['Charged Off', 'Fully Paid'],
                                 mode='classification',
                                 random_state=42)

i = 37
```

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```
def predict_fn(x):
    # Get the raw predictions from your model
    preds = ANN_model.predict(x)

    if len(preds.shape) == 1 or preds.shape[1] == 1:
        # Return probabilities for both classes
        return np.column_stack([1-preds, preds])
    else:
        return preds

exp = explainer.explain_instance(x_test_2[i], predict_fn, num_features=50)

print(y_test_2[i])

exp.show_in_notebook(show_table=True)

exp.save_to_file('explanation_1.html')


# Check calculations
exp_weight = [x_test_2[1] for x_test_2 in exp.as_map()[1]]

print(np.mean(y_test_2) + sum(exp_weight))
print(y_test_2[0])


def return_weights(exp):

    """Get weights from LIME explanation object"""

    exp_list = exp.as_map()[1]
    exp_list = sorted(exp_list, key=lambda x_test: x_test[0])
    exp_weight = [x_test[1] for x_test in exp_list]

    return exp_weight

feature_names.remove('loan_status')
print(len(feature_names))

weights = []

exp = explainer.explain_instance(x_test_2[0], predict_fn, num_features=15)
```

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```
# Iterate over first 100 rows
for i in range(0,100):

    # Get explanation
    exp = explainer.explain_instance(x_test_2[0], predict_fn, num_features=146)

    # Get weights
    exp_weight = return_weights(exp)
    weights.append(exp_weight)

exp.show_in_notebook(show_table=True)

# Create DataFrame
lime_weights = pd.DataFrame(data=weights, columns = feature_names)

# lime_weights = pd.DataFrame(data=weights, columns = feature_names)

masker = shap.maskers.Independent(x_train_2[0:100])

explainer = shap.Explainer(ANN_model, masker)
shap_values = explainer(x_train_2[0:100])

# Replace SHAP values with LIME weights
shap_placeholder = explainer(x_train_2[0:100])
shap_placeholder.values = np.array(lime_weights)

explanation = shap.Explanation(
    values=shap_placeholder.values,
    base_values=shap_placeholder.base_values,
    data=shap_placeholder.data,
    feature_names=feature_names
)

plt.xlabel("SHAP Values")
plt.title('SHAP Values of top 9 features and the remaining features for single instance')
shap.plots.waterfall(shap_values[0],show=False)
plt.show()

plt.xlabel("SHAP Values")
plt.title('SHAP Values of top 9 features and the remaining features for single instance')
shap.plots.waterfall(shap_values[4],show=False)
plt.show()
```

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```
plt.xlabel("Mean LIME Weight")  
plt.title('Mean LIME Weight of top 9 features and the remaining features for single instance')  
shap.plots.waterfall(explanation[0],show=False)  
plt.show()
```

```
plt.xlabel("Mean LIME Weight")  
plt.title('Mean LIME Weight of top 9 features and the remaining features for single instance')  
shap.plots.waterfall(explanation[4],show=False)  
plt.show()
```